tropical: Quickselect

2026-10-25T12:45:32

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `bd73c304e6b9`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.7137	[0.705, 0.722]	0.0107	5	0
adaptive	pointer	0.01	0.7294	[0.716, 0.741]	0.017	5	0
adaptive	pointer	0.1	0.7264	[0.714, 0.736]	0.0147	5	0
adaptive	pointer	1	0.7218	[0.708, 0.736]	0.0185	5	0
adaptive	pointer	10	0.7308	[0.717, 0.742]	0.016	5	0
adaptive	pointer	100	0.7228	[0.711, 0.735]	0.016	5	0
adaptive	pointer	1000	0.733	[0.728, 0.738]	0.00699	5	0
adaptive	pointer	1e+04	0.7338	[0.722, 0.746]	0.0143	5	0
adaptive	seq2seq	0	0.6747	[0.658, 0.704]	0.0326	5	0
adaptive	seq2seq	0.01	0.6878	[0.669, 0.703]	0.0212	5	0
adaptive	seq2seq	0.1	0.6799	[0.655, 0.699]	0.0283	5	0
adaptive	seq2seq	1	0.6916	[0.684, 0.7]	0.00997	5	0
adaptive	seq2seq	10	0.7004	[0.683, 0.716]	0.0213	5	0
adaptive	seq2seq	100	0.6857	[0.676, 0.701]	0.0164	5	0
adaptive	seq2seq	1000	0.6545	[0.63, 0.678]	0.0292	5	0
adaptive	seq2seq	1e+04	0.5728	[0.539, 0.6]	0.0412	5	0
tropical	pointer	0	0.7494	[0.721, 0.783]	0.0436	5	0
tropical	pointer	0.01	0.7304	[0.72, 0.741]	0.0138	5	0
tropical	pointer	0.1	0.7359	[0.726, 0.744]	0.0119	5	0
tropical	pointer	1	0.7312	[0.722, 0.737]	0.0106	5	0
tropical	pointer	10	0.7364	[0.716, 0.754]	0.024	5	0
tropical	pointer	100	0.7411	[0.738, 0.747]	0.00605	5	0
tropical	pointer	1000	0.7241	[0.693, 0.75]	0.0378	5	0
tropical	pointer	1e+04	0.7111	[0.676, 0.733]	0.0393	5	0
tropical	seq2seq	0	0.5472	[0.488, 0.61]	0.0809	5	0
tropical	seq2seq	0.01	0.5201	[0.422, 0.618]	0.131	5	0
tropical	seq2seq	0.1	0.4432	[0.355, 0.536]	0.116	5	0
tropical	seq2seq	1	0.4799	[0.379, 0.568]	0.119	5	0
tropical	seq2seq	10	0.4399	[0.34, 0.552]	0.137	5	0
tropical	seq2seq	100	0.3488	[0.285, 0.397]	0.073	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.4323	[0.317, 0.574]	0.162	5	0
tropical	seq2seq	1e+04	0.6917	[0.677, 0.707]	0.0196	5	0
vanilla	pointer	0	0.6521	[0.643, 0.665]	0.0133	5	0
vanilla	pointer	0.01	0.6254	[0.613, 0.642]	0.018	5	0
vanilla	pointer	0.1	0.6298	[0.615, 0.641]	0.0169	5	0
vanilla	pointer	1	0.6258	[0.621, 0.632]	0.00767	5	0
vanilla	pointer	10	0.6254	[0.62, 0.63]	0.00612	5	0
vanilla	pointer	100	0.6226	[0.617, 0.629]	0.00832	5	0
vanilla	pointer	1000	0.6322	[0.621, 0.644]	0.016	5	0
vanilla	pointer	1e+04	0.6546	[0.641, 0.67]	0.0174	5	0
vanilla	seq2seq	0	0.469	[0.434, 0.505]	0.047	5	0
vanilla	seq2seq	0.01	0.4898	[0.478, 0.504]	0.0173	5	0
vanilla	seq2seq	0.1	0.4918	[0.475, 0.514]	0.0249	5	0
vanilla	seq2seq	1	0.4884	[0.437, 0.525]	0.0532	5	0
vanilla	seq2seq	10	0.5135	[0.482, 0.545]	0.0406	5	0
vanilla	seq2seq	100	0.4909	[0.461, 0.52]	0.0396	5	0
vanilla	seq2seq	1000	0.4881	[0.46, 0.516]	0.0358	5	0
vanilla	seq2seq	1e+04	0.3965	[0.376, 0.418]	0.0275	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.01571	[-0.00995, 0.0414]	0.3955	
adaptive	pointer	0.1	0.01271	[-0.0129, 0.0384]	0.6103	
adaptive	pointer	1	0.008079	[-0.0176, 0.0337]	0.9176	
adaptive	pointer	10	0.01703	[-0.00864, 0.0427]	0.3163	
adaptive	pointer	100	0.009039	[-0.0166, 0.0347]	0.8691	
adaptive	pointer	1000	0.01926	[-0.00641, 0.0449]	0.2082	
adaptive	pointer	1e+04	0.0201	[-0.00556, 0.0458]	0.1755	
adaptive	seq2seq	0.01	0.01318	[-0.0334, 0.0597]	0.9491	
adaptive	seq2seq	0.1	0.005239	[-0.0413, 0.0518]	0.9998	
adaptive	seq2seq	1	0.01694	[-0.0296, 0.0635]	0.8520	
adaptive	seq2seq	10	0.02572	[-0.0209, 0.0723]	0.5010	
adaptive	seq2seq	100	0.01103	[-0.0355, 0.0576]	0.9791	
adaptive	seq2seq	1000	-0.02019	[-0.0668, 0.0264]	0.7307	
adaptive	seq2seq	1e+04	-0.1019	[-0.148, -0.0553]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.01897	[-0.0665, 0.0286]	0.7937	
tropical	pointer	0.1	-0.0135	[-0.0611, 0.0341]	0.9484	
tropical	pointer	1	-0.01819	[-0.0657, 0.0294]	0.8219	
tropical	pointer	10	-0.01303	[-0.0606, 0.0345]	0.9563	
tropical	pointer	100	-0.008305	[-0.0559, 0.0392]	0.9962	
tropical	pointer	1000	-0.02535	[-0.0729, 0.0222]	0.5373	
tropical	pointer	1e+04	-0.03835	[-0.0859, 0.0092]	0.1552	
tropical	seq2seq	0.01	-0.02709	[-0.224, 0.17]	0.9991	
tropical	seq2seq	0.1	-0.1039	[-0.301, 0.0932]	0.5485	
tropical	seq2seq	1	-0.06722	[-0.264, 0.13]	0.8848	
tropical	seq2seq	10	-0.1072	[-0.304, 0.0899]	0.5163	
tropical	seq2seq	100	-0.1984	[-0.395, -0.00125]	0.0480	*
tropical	seq2seq	1000	-0.1149	[-0.312, 0.0822]	0.4456	
tropical	seq2seq	1e+04	0.1446	[-0.0525, 0.342]	0.2265	
vanilla	pointer	0.01	-0.02673	[-0.0508, -0.0027]	0.0240	*
vanilla	pointer	0.1	-0.02235	[-0.0464, 0.00167]	0.0765	
vanilla	pointer	1	-0.0264	[-0.0504, -0.00237]	0.0263	*

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
vanilla	pointer	10	-0.02679	[-0.0508, -0.00277]	0.0237	*
vanilla	pointer	100	-0.02954	[-0.0536, -0.00551]	0.0107	*
vanilla	pointer	1000	-0.01994	[-0.044, 0.00409]	0.1367	
vanilla	pointer	1e+04	0.002412	[-0.0216, 0.0264]	0.9999	
vanilla	seq2seq	0.01	0.02076	[-0.0446, 0.0862]	0.9145	
vanilla	seq2seq	0.1	0.02276	[-0.0426, 0.0882]	0.8751	
vanilla	seq2seq	1	0.01934	[-0.0461, 0.0847]	0.9376	
vanilla	seq2seq	10	0.04445	[-0.0209, 0.11]	0.2936	
vanilla	seq2seq	100	0.02185	[-0.0435, 0.0873]	0.8939	
vanilla	seq2seq	1000	0.01907	[-0.0463, 0.0845]	0.9415	
vanilla	seq2seq	1e+04	-0.07253	[-0.138, -0.00713]	0.0247	*

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	9.724×10^{-4}	$[-8.18 \times 10^{-4}, 0.0033]$	0.4352	0.124	0.3605
adaptive	seq2seq	-0.01436	[-0.0181, -0.0107]	$< 10^{-4}$	-0.467	0.0006
tropical	pointer	-0.00257	[-0.00951, 0.00268]	0.2040	-0.0629	0.6457
tropical	seq2seq	0.01293	[-0.0101, 0.0285]	0.3020	0.109	0.4095
vanilla	pointer	0.003182	$[-6.68 \times 10^{-4}, 0.00793]$	0.0178	0.242	0.0764
vanilla	seq2seq	-0.01017	[-0.0138, -0.00686]	0.0103	-0.265	0.0502

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.09726	[0.0367, 0.158]	0.1114	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.105	[0.0909, 0.119]	0.0004	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1061	[0.0816, 0.131]	0.0033	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1055	[0.0925, 0.118]	0.0003	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.111	[0.0837, 0.138]	0.0039	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1185	[0.106, 0.131]	0.0002	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.09185	[0.0345, 0.149]	0.1114	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.05649	[0.00835, 0.105]	0.2243	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.07814	[-0.0794, 0.236]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0303	[-0.118, 0.179]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.04853	[-0.216, 0.119]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.008412	[-0.21, 0.193]	1.0000	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.07355	[-0.288, 0.141]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.1421	[-0.259, -0.025]	0.2243	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.05579	[-0.27, 0.158]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.2953	[0.258, 0.333]	0.0004	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.06158	[0.0327, 0.0904]	0.0057	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.104	[0.0907, 0.117]	0.0004	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.09665	[0.0869, 0.106]	0.0002	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.09605	[0.0758, 0.116]	0.0024	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.1054	[0.0833, 0.127]	0.0024	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.1002	[0.0711, 0.129]	0.0042	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.1008	[0.0763, 0.125]	0.0034	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.07927	[0.0512, 0.107]	0.0057	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.2056	[0.12, 0.292]	0.0057	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.1981	[0.178, 0.219]	0.0002	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.1881	[0.147, 0.23]	0.0025	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.2032	[0.129, 0.278]	0.0057	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.1869	[0.133, 0.241]	0.0042	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.1948	[0.145, 0.244]	0.0036	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.1664	[0.121, 0.212]	0.0042	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.1763	[0.128, 0.225]	0.0042	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.03906	[0.00426, 0.0739]	0.1850	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0416	$[-8.78 \times 10^{-4}, 0.0841]$	0.2063	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.04654	$[-5.09 \times 10^{-4}, 0.0936]$	0.2063	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.03021	[0.00535, 0.0551]	0.1850	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.03037	[-0.0116, 0.0723]	0.2297	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.03708	[0.00834, 0.0658]	0.1850	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.07851	[0.0364, 0.121]	0.0859	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1611	[0.111, 0.211]	0.0160	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.2022	[0.0602, 0.344]	0.1511	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.2104	[0.0445, 0.376]	0.1850	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.2927	[0.156, 0.429]	0.0563	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.2513	[0.113, 0.39]	0.0859	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2965	[0.136, 0.457]	0.0859	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.3923	[0.304, 0.48]	0.0057	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2918	[0.117, 0.467]	0.0981	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.01931	[-0.0322, 0.0709]	0.3570	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1831	[0.13, 0.236]	0.0132	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1356	[0.0974, 0.174]	0.0126	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.138	[0.103, 0.173]	0.0087	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1374	[0.077, 0.198]	0.0513	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1119	[0.0611, 0.163]	0.0544	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1317	[0.0771, 0.186]	0.0440	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1441	[0.101, 0.187]	0.0141	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2581	[0.221, 0.296]	0.0011	*

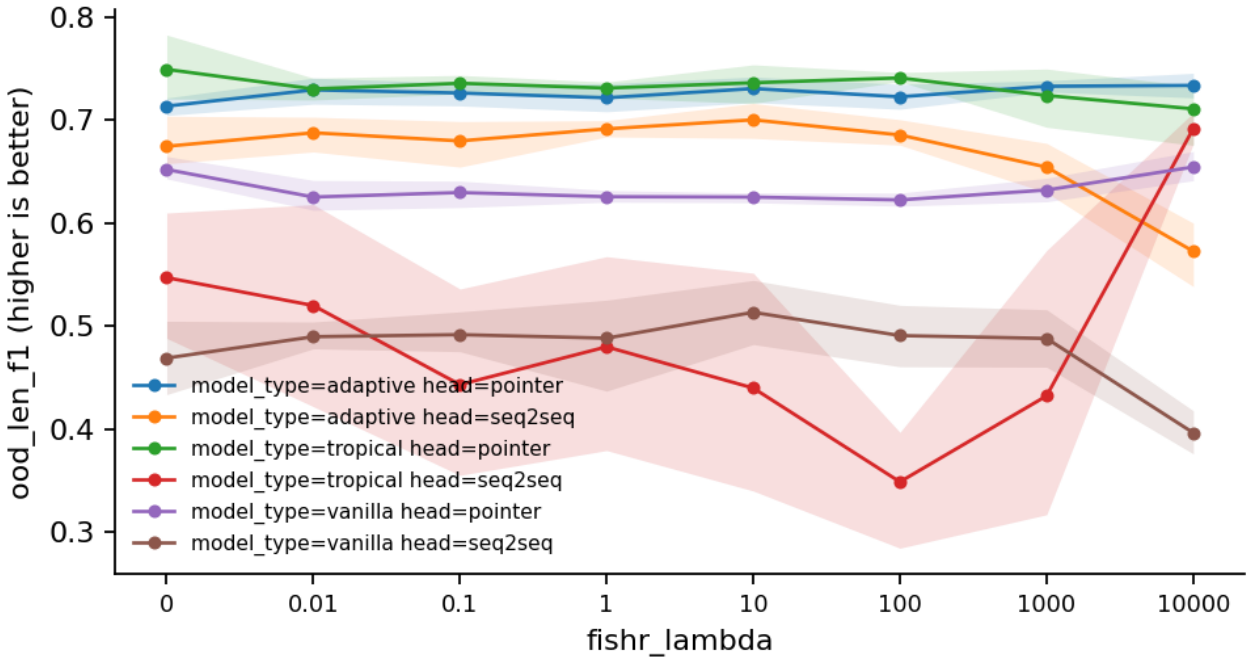


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

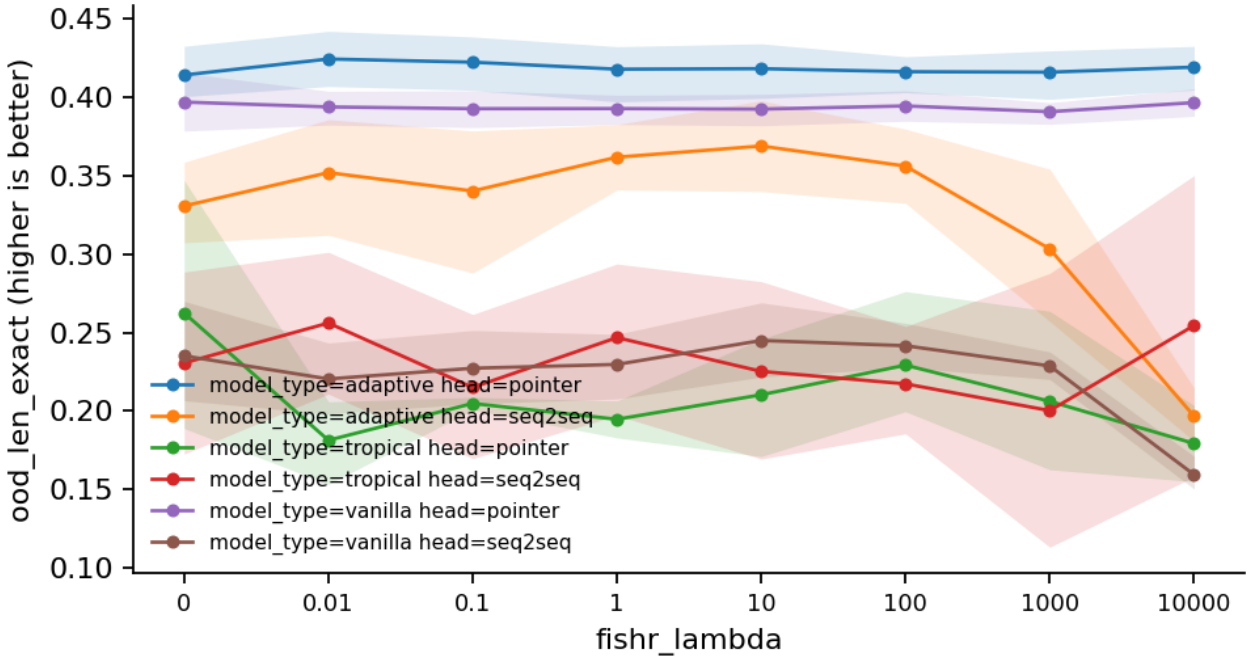


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4144	[0.401, 0.433]	0.0204	5	0
adaptive	pointer	0.01	0.4246	[0.407, 0.442]	0.0213	5	0
adaptive	pointer	0.1	0.4225	[0.405, 0.439]	0.0221	5	0
adaptive	pointer	1	0.4181	[0.397, 0.432]	0.022	5	0
adaptive	pointer	10	0.4184	[0.4, 0.434]	0.0205	5	0
adaptive	pointer	100	0.4165	[0.403, 0.426]	0.0152	5	0
adaptive	pointer	1000	0.4162	[0.399, 0.43]	0.02	5	0
adaptive	pointer	1e+04	0.4194	[0.405, 0.433]	0.0188	5	0
adaptive	seq2seq	0	0.3308	[0.307, 0.359]	0.0346	5	0
adaptive	seq2seq	0.01	0.3521	[0.312, 0.386]	0.047	5	0
adaptive	seq2seq	0.1	0.3404	[0.288, 0.379]	0.0575	5	0
adaptive	seq2seq	1	0.3619	[0.341, 0.383]	0.0266	5	0
adaptive	seq2seq	10	0.3691	[0.34, 0.398]	0.04	5	0
adaptive	seq2seq	100	0.3563	[0.333, 0.38]	0.031	5	0
adaptive	seq2seq	1000	0.3034	[0.257, 0.354]	0.0599	5	0
adaptive	seq2seq	1e+04	0.1967	[0.181, 0.215]	0.0203	5	0
tropical	pointer	0	0.2626	[0.189, 0.347]	0.106	5	0
tropical	pointer	0.01	0.1814	[0.153, 0.206]	0.0345	5	0
tropical	pointer	0.1	0.2048	[0.201, 0.209]	0.00521	5	0
tropical	pointer	1	0.1947	[0.183, 0.207]	0.0156	5	0
tropical	pointer	10	0.2104	[0.171, 0.246]	0.0453	5	0
tropical	pointer	100	0.2293	[0.2, 0.276]	0.0532	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.2062	[0.163, 0.264]	0.067	5	0
tropical	pointer	1e+04	0.1793	[0.155, 0.204]	0.0335	5	0
tropical	seq2seq	0	0.2307	[0.173, 0.289]	0.0735	5	0
tropical	seq2seq	0.01	0.2561	[0.211, 0.301]	0.0609	5	0
tropical	seq2seq	0.1	0.215	[0.17, 0.262]	0.0618	5	0
tropical	seq2seq	1	0.2468	[0.197, 0.294]	0.0614	5	0
tropical	seq2seq	10	0.2253	[0.169, 0.283]	0.0731	5	0
tropical	seq2seq	100	0.2173	[0.186, 0.254]	0.0421	5	0
tropical	seq2seq	1000	0.2003	[0.113, 0.288]	0.111	5	0
tropical	seq2seq	1e+04	0.2545	[0.158, 0.35]	0.128	5	0
vanilla	pointer	0	0.3971	[0.379, 0.416]	0.0246	5	0
vanilla	pointer	0.01	0.394	[0.383, 0.404]	0.0136	5	0
vanilla	pointer	0.1	0.3929	[0.381, 0.404]	0.015	5	0
vanilla	pointer	1	0.3929	[0.383, 0.401]	0.0118	5	0
vanilla	pointer	10	0.3926	[0.382, 0.402]	0.0133	5	0
vanilla	pointer	100	0.3946	[0.385, 0.404]	0.0124	5	0
vanilla	pointer	1000	0.3909	[0.383, 0.397]	0.00897	5	0
vanilla	pointer	1e+04	0.3968	[0.388, 0.405]	0.0111	5	0
vanilla	seq2seq	0	0.2352	[0.207, 0.27]	0.0397	5	0
vanilla	seq2seq	0.01	0.2205	[0.198, 0.243]	0.0292	5	0
vanilla	seq2seq	0.1	0.2273	[0.205, 0.251]	0.0303	5	0
vanilla	seq2seq	1	0.2297	[0.208, 0.249]	0.0248	5	0
vanilla	seq2seq	10	0.245	[0.222, 0.269]	0.0308	5	0
vanilla	seq2seq	100	0.2417	[0.228, 0.256]	0.0185	5	0
vanilla	seq2seq	1000	0.2288	[0.22, 0.238]	0.0116	5	0
vanilla	seq2seq	1e+04	0.1594	[0.15, 0.171]	0.0134	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.0102	[-0.025, 0.0454]	0.9430	
adaptive	pointer	0.1	0.008167	[-0.027, 0.0434]	0.9812	
adaptive	pointer	1	0.0037	[-0.0315, 0.0389]	0.9998	
adaptive	pointer	10	0.004033	[-0.0312, 0.0392]	0.9997	
adaptive	pointer	100	0.0021	[-0.0331, 0.0373]	1.0000	
adaptive	pointer	1000	0.0018	[-0.0334, 0.037]	1.0000	
adaptive	pointer	1e+04	0.005	[-0.0302, 0.0402]	0.9989	
adaptive	seq2seq	0.01	0.0213	[-0.0517, 0.0943]	0.9413	
adaptive	seq2seq	0.1	0.009567	[-0.0635, 0.0826]	0.9994	
adaptive	seq2seq	1	0.03113	[-0.0419, 0.104]	0.7440	
adaptive	seq2seq	10	0.03827	[-0.0348, 0.111]	0.5544	
adaptive	seq2seq	100	0.02547	[-0.0476, 0.0985]	0.8740	
adaptive	seq2seq	1000	-0.0274	[-0.1, 0.0456]	0.8337	
adaptive	seq2seq	1e+04	-0.1341	[-0.207, -0.061]	0.0001	*
tropical	pointer	0.01	-0.0812	[-0.175, 0.0127]	0.1128	
tropical	pointer	0.1	-0.05773	[-0.152, 0.0362]	0.3916	
tropical	pointer	1	-0.06783	[-0.162, 0.0261]	0.2399	
tropical	pointer	10	-0.0522	[-0.146, 0.0417]	0.4947	
tropical	pointer	100	-0.0333	[-0.127, 0.0606]	0.8658	
tropical	pointer	1000	-0.05637	[-0.15, 0.0376]	0.4159	
tropical	pointer	1e+04	-0.08327	[-0.177, 0.0107]	0.0994	
tropical	seq2seq	0.01	0.02547	[-0.116, 0.167]	0.9956	
tropical	seq2seq	0.1	-0.01567	[-0.157, 0.126]	0.9998	

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	seq2seq	1	0.01617	[-0.126, 0.158]	0.9997	
tropical	seq2seq	10	-0.005367	[-0.147, 0.136]	1.0000	
tropical	seq2seq	100	-0.01333	[-0.155, 0.128]	0.9999	
tropical	seq2seq	1000	-0.0304	[-0.172, 0.111]	0.9877	
tropical	seq2seq	1e+04	0.02387	[-0.118, 0.166]	0.9970	
vanilla	pointer	0.01	-0.0031	[-0.0285, 0.0223]	0.9996	
vanilla	pointer	0.1	-0.004233	[-0.0296, 0.0211]	0.9971	
vanilla	pointer	1	-0.004167	[-0.0295, 0.0212]	0.9974	
vanilla	pointer	10	-0.0045	[-0.0299, 0.0209]	0.9959	
vanilla	pointer	100	-0.002467	[-0.0278, 0.0229]	0.9999	
vanilla	pointer	1000	-0.006167	[-0.0315, 0.0192]	0.9760	
vanilla	pointer	1e+04	-3.333×10^{-4}	[-0.0257, 0.025]	1.0000	
vanilla	seq2seq	0.01	-0.01473	[-0.0608, 0.0313]	0.9117	
vanilla	seq2seq	0.1	-0.0079	[-0.054, 0.0382]	0.9966	
vanilla	seq2seq	1	-0.005467	[-0.0515, 0.0406]	0.9997	
vanilla	seq2seq	10	0.0098	[-0.0363, 0.0559]	0.9883	
vanilla	seq2seq	100	0.006467	[-0.0396, 0.0525]	0.9990	
vanilla	seq2seq	1000	-0.006433	[-0.0525, 0.0396]	0.9991	
vanilla	seq2seq	1e+04	-0.0758	[-0.122, -0.0297]	0.0005	*

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.001069	[-0.00185, -4.86×10^{-4}]	0.5018	-0.103	0.4449
adaptive	seq2seq	-0.01949	[-0.0266, -0.0139]	0.0003	-0.379	0.0036
tropical	pointer	0.00111	[-0.00458, 0.0101]	0.7536	0.0438	0.7536
tropical	seq2seq	-0.002277	[-0.0163, 0.012]	0.7331	-0.0667	0.6173
vanilla	pointer	2.190×10^{-4}	[-0.00104, 0.00143]	0.8242	0	1.0000
vanilla	seq2seq	-0.006012	[-0.00922, -0.00287]	0.0380	-0.21	0.1114

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.1345	[-0.253, -0.0156]	0.3133	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.2126	[-0.269, -0.156]	0.0062	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.188	[-0.204, -0.172]	$< 10^{-4}$	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.1982	[-0.225, -0.171]	0.0005	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.1822	[-0.242, -0.122]	0.0128	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.1654	[-0.225, -0.106]	0.0164	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.1847	[-0.27, -0.0999]	0.0378	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.2175	[-0.264, -0.171]	0.0028	*

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.004533	[-0.132, 0.123]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.03567	[-0.0516, 0.123]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.0123	[-0.125, 0.1]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.0171	[-0.0621, 0.0963]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.0197	[-0.145, 0.106]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.02433	[-0.0947, 0.0461]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0285	[-0.18, 0.123]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.09513	[-0.0613, 0.252]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01727	[0.0015, 0.033]	0.2685	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.03057	[-0.00445, 0.0656]	0.3632	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02967	[-0.00667, 0.066]	0.3632	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.02513	[-0.0076, 0.0579]	0.3632	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0258	[-0.0047, 0.0563]	0.3632	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02183	[-0.00348, 0.0471]	0.3632	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.02523	[-0.00179, 0.0523]	0.3632	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.0226	[0.00418, 0.041]	0.2437	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.0956	[0.0203, 0.171]	0.2437	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.1316	[0.0876, 0.176]	0.0156	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.1131	[0.0569, 0.169]	0.0602	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.1322	[0.0883, 0.176]	0.0156	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.1241	[0.0842, 0.164]	0.0148	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.1146	[0.0916, 0.138]	0.0025	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.07463	[0.0139, 0.135]	0.2437	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.03733	[0.0118, 0.0629]	0.1697	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.08357	[0.0312, 0.136]	0.1716	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.07247	[-0.00192, 0.147]	0.5114	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.08217	$[-6.55 \times 10^{-4}, 0.165]$	0.5114	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.05613	[0.00192, 0.11]	0.4977	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.04933	[-0.0171, 0.116]	0.8645	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.0602	[0.00989, 0.111]	0.3811	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.1128	[0.0171, 0.208]	0.3811	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.2226	[0.197, 0.249]	0.0004	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0319	[-0.146, 0.21]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.07477	[-0.135, -0.0141]	0.3743	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01017	[-0.0913, 0.0709]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.0521	[-0.129, 0.0251]	0.9388	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.01493	[-0.125, 0.0955]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01193	[-0.0396, 0.0634]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.005933	[-0.0929, 0.105]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.07523	[-0.23, 0.0793]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1619	[0.116, 0.208]	0.0100	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1735	[0.143, 0.204]	0.0019	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1656	[0.131, 0.2]	0.0034	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1632	[0.128, 0.199]	0.0038	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1476	[0.116, 0.18]	0.0038	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.153	[0.129, 0.177]	0.0013	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1622	[0.148, 0.176]	0.0001	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2374	[0.211, 0.263]	0.0003	*

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9957	[0.995, 0.997]	0.00113	5	0
adaptive	pointer	0.01	0.9957	[0.995, 0.996]	6.68×10^{-4}	5	0
adaptive	pointer	0.1	0.9958	[0.995, 0.996]	5.39×10^{-4}	5	0
adaptive	pointer	1	0.9958	[0.995, 0.996]	4.69×10^{-4}	5	0
adaptive	pointer	10	0.9957	[0.995, 0.996]	5.12×10^{-4}	5	0
adaptive	pointer	100	0.9957	[0.995, 0.996]	5.03×10^{-4}	5	0
adaptive	pointer	1000	0.996	[0.996, 0.996]	4.11×10^{-4}	5	0
adaptive	pointer	1e+04	0.996	[0.996, 0.996]	2.87×10^{-4}	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.9336	[0.933, 0.934]	0.0012	5	0
adaptive	seq2seq	0.01	0.9331	[0.932, 0.934]	0.00161	5	0
adaptive	seq2seq	0.1	0.9329	[0.932, 0.934]	0.00147	5	0
adaptive	seq2seq	1	0.9331	[0.932, 0.934]	0.00158	5	0
adaptive	seq2seq	10	0.9331	[0.932, 0.934]	0.0013	5	0
adaptive	seq2seq	100	0.9321	[0.93, 0.935]	0.00316	5	0
adaptive	seq2seq	1000	0.9277	[0.926, 0.929]	0.00218	5	0
adaptive	seq2seq	1e+04	0.9178	[0.915, 0.921]	0.00403	5	0
tropical	pointer	0	0.9329	[0.912, 0.951]	0.0262	5	0
tropical	pointer	0.01	0.9362	[0.919, 0.952]	0.0226	5	0
tropical	pointer	0.1	0.9371	[0.918, 0.954]	0.0239	5	0
tropical	pointer	1	0.9347	[0.917, 0.951]	0.0218	5	0
tropical	pointer	10	0.9356	[0.916, 0.951]	0.0217	5	0
tropical	pointer	100	0.9352	[0.918, 0.952]	0.0223	5	0
tropical	pointer	1000	0.9376	[0.922, 0.954]	0.0208	5	0
tropical	pointer	1e+04	0.9363	[0.919, 0.952]	0.0213	5	0
tropical	seq2seq	0	0.7523	[0.742, 0.763]	0.0138	5	0
tropical	seq2seq	0.01	0.7487	[0.742, 0.756]	0.00875	5	0
tropical	seq2seq	0.1	0.7529	[0.748, 0.758]	0.00591	5	0
tropical	seq2seq	1	0.7508	[0.744, 0.757]	0.00856	5	0
tropical	seq2seq	10	0.7564	[0.751, 0.761]	0.00706	5	0
tropical	seq2seq	100	0.7429	[0.722, 0.759]	0.0239	5	0
tropical	seq2seq	1000	0.7238	[0.694, 0.75]	0.0362	5	0
tropical	seq2seq	1e+04	0.6796	[0.629, 0.719]	0.0581	5	0
vanilla	pointer	0	0.9949	[0.995, 0.995]	4.49×10^{-4}	5	0
vanilla	pointer	0.01	0.9954	[0.995, 0.996]	7.59×10^{-4}	5	0
vanilla	pointer	0.1	0.9953	[0.995, 0.996]	6.32×10^{-4}	5	0
vanilla	pointer	1	0.9954	[0.995, 0.996]	7.46×10^{-4}	5	0
vanilla	pointer	10	0.9955	[0.995, 0.996]	6.25×10^{-4}	5	0
vanilla	pointer	100	0.9955	[0.995, 0.996]	6.06×10^{-4}	5	0
vanilla	pointer	1000	0.9955	[0.995, 0.996]	5.64×10^{-4}	5	0
vanilla	pointer	1e+04	0.9956	[0.995, 0.996]	4.32×10^{-4}	5	0
vanilla	seq2seq	0	0.9349	[0.933, 0.937]	0.00286	5	0
vanilla	seq2seq	0.01	0.9348	[0.932, 0.938]	0.00347	5	0
vanilla	seq2seq	0.1	0.9347	[0.932, 0.937]	0.00316	5	0
vanilla	seq2seq	1	0.9352	[0.932, 0.938]	0.00371	5	0
vanilla	seq2seq	10	0.935	[0.933, 0.938]	0.00325	5	0
vanilla	seq2seq	100	0.934	[0.931, 0.937]	0.00404	5	0
vanilla	seq2seq	1000	0.9258	[0.924, 0.928]	0.00253	5	0
vanilla	seq2seq	1e+04	0.9149	[0.913, 0.917]	0.00292	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	1.329×10^{-4}	[-0.00112, 0.00116]	1.0000
adaptive	pointer	0.1	-9.965×10^{-5}	$[-7.91 \times 10^{-4}, 0.00108]$	1.0000
adaptive	pointer	1	-3.264×10^{-5}	$[-8.38 \times 10^{-4}, 0.00104]$	1.0000
adaptive	pointer	10	-9.965×10^{-5}	$[-7.91 \times 10^{-4}, 0.00101]$	1.0000
adaptive	pointer	100	1.681×10^{-4}	[-0.00109, 0.00115]	1.0000
adaptive	pointer	1000	3.705×10^{-4}	$[-7.50 \times 10^{-4}, 0.0014]$	1.0000
adaptive	pointer	1e+04	3.699×10^{-4}	$[-4.52 \times 10^{-4}, 0.00117]$	1.0000
adaptive	seq2seq	0.01	2.385×10^{-5}	$[-0.00145, 3.07 \times 10^{-4}]$	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	0.1	-5.365×10^{-4}	[-0.00122, -1.87×10^{-4}]	0.6250
adaptive	seq2seq	1	7.383×10^{-5}	[-0.00153, 3.18×10^{-4}]	1.0000
adaptive	seq2seq	10	-2.941×10^{-4}	[-0.00103, -5.75×10^{-5}]	0.7500
adaptive	seq2seq	100	-0.001759	[-0.0033, 1.75×10^{-4}]	0.9375
adaptive	seq2seq	1000	-0.005248	[-0.00747, -0.0049]	0.4375
adaptive	seq2seq	1e+04	-0.01547	[-0.0185, -0.013]	0.4375
tropical	pointer	0.01	0.002545	[-4.79×10^{-5}, 0.00692]	1.0000
tropical	pointer	0.1	0.005462	[7.40×10^{-4}, 0.00705]	1.0000
tropical	pointer	1	7.079×10^{-5}	[-0.00179, 0.00596]	1.0000
tropical	pointer	10	0.002024	[-0.00103, 0.00648]	1.0000
tropical	pointer	100	9.750×10^{-4}	[-0.00133, 0.006]	1.0000
tropical	pointer	1000	0.00233	[1.19×10^{-4}, 0.01]	1.0000
tropical	pointer	1e+04	0.007318	[-0.00308, 0.00818]	1.0000
tropical	seq2seq	0.01	-0.003136	[-0.00801, 8.97×10^{-4}]	1.0000
tropical	seq2seq	0.1	0.00243	[-0.00604, 0.00724]	1.0000
tropical	seq2seq	1	0.001091	[-0.00632, 0.00287]	1.0000
tropical	seq2seq	10	0.007181	[-0.00405, 0.0121]	1.0000
tropical	seq2seq	100	-0.002194	[-0.0258, 0.00283]	1.0000
tropical	seq2seq	1000	-0.01786	[-0.0527, -0.00899]	0.7500
tropical	seq2seq	1e+04	-0.04512	[-0.114, -0.0412]	0.4375
vanilla	pointer	0.01	4.980×10^{-4}	[5.55×10^{-6}, 9.42×10^{-4}]	0.9375
vanilla	pointer	0.1	2.643×10^{-4}	[5.41×10^{-6}, 8.92×10^{-4}]	0.9375
vanilla	pointer	1	4.980×10^{-4}	[-2.15×10^{-5}, 9.73×10^{-4}]	0.9375
vanilla	pointer	10	5.033×10^{-4}	[2.19×10^{-4}, 0.00105]	0.6250
vanilla	pointer	100	6.040×10^{-4}	[1.49×10^{-4}, 0.00114]	0.6250
vanilla	pointer	1000	4.316×10^{-4}	[2.59×10^{-4}, 9.29×10^{-4}]	0.4375
vanilla	pointer	1e+04	6.974×10^{-4}	[3.80×10^{-4}, 0.00104]	0.4375
vanilla	seq2seq	0.01	1.940×10^{-4}	[-0.00109, 7.49×10^{-4}]	1.0000
vanilla	seq2seq	0.1	-2.173×10^{-4}	[-9.58×10^{-4}, 6.51×10^{-4}]	1.0000
vanilla	seq2seq	1	8.150×10^{-4}	[-7.24×10^{-4}, 0.00126]	1.0000
vanilla	seq2seq	10	4.467×10^{-5}	[-8.28×10^{-4}, 9.10×10^{-4}]	1.0000
vanilla	seq2seq	100	-0.001157	[-0.00268, 0.00127]	1.0000
vanilla	seq2seq	1000	-0.008846	[-0.00998, -0.0082]	0.4375
vanilla	seq2seq	1e+04	-0.0202	[-0.0214, -0.0185]	0.4375

Table 11: `ood_val_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.2115	[-0.00569, 0.553]	0.2225	0.158	0.2456
adaptive	seq2seq	-0.6493	[-0.731, -0.587]	$< 10^{-4}$	-0.528	0.0002
tropical	pointer	-0.004243	[-0.254, 0.133]	0.9807	0.00571	0.9768
tropical	seq2seq	-0.4456	[-0.564, -0.424]	0.0073	-0.322	0.0152
vanilla	pointer	0.1429	[-0.03, 0.554]	0.4129	0.118	0.3841
vanilla	seq2seq	-0.6705	[-0.741, -0.587]	$< 10^{-4}$	-0.562	0.0002

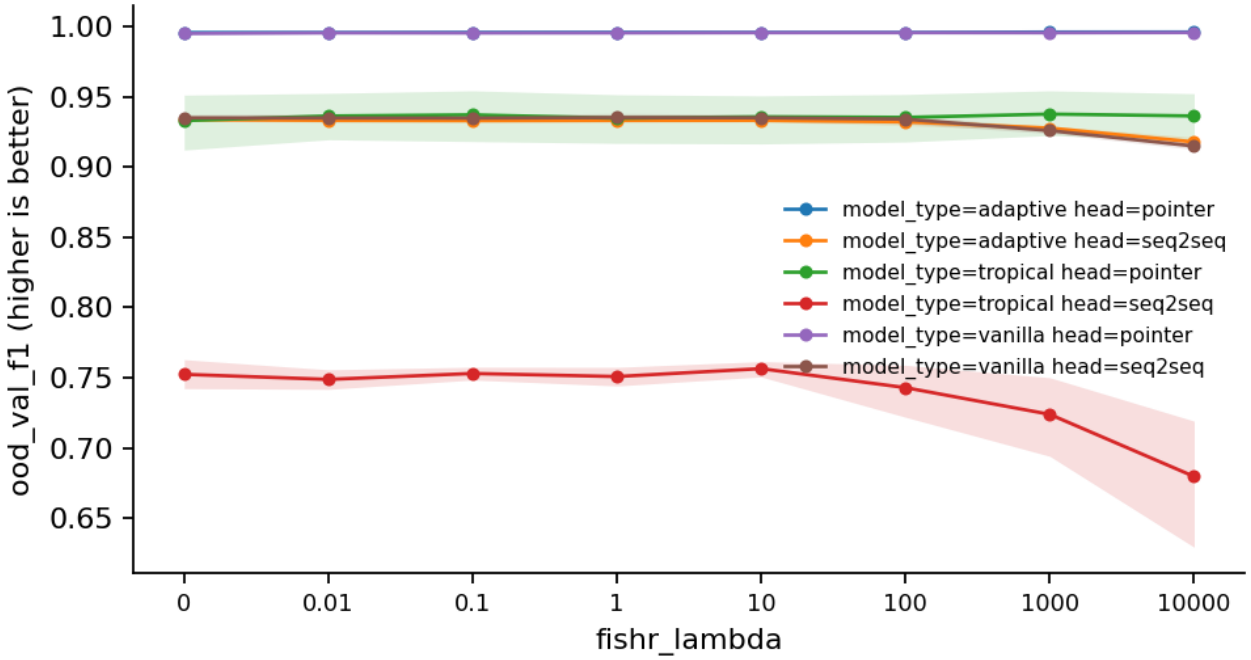


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.06198	[-0.0948, -0.0292]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.05915	[-0.0881, -0.0302]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.05826	[-0.0887, -0.0278]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.06071	[-0.0886, -0.0328]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.05988	[-0.0875, -0.0322]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.0603	[-0.0887, -0.0319]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.05789	[-0.0844, -0.0314]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.05936	[-0.0864, -0.0323]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.1826	[-0.201, -0.164]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.1861	[-0.199, -0.173]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.1818	[-0.191, -0.173]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.1844	[-0.197, -0.172]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.1786	[-0.19, -0.167]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.1911	[-0.221, -0.162]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.202	[-0.247, -0.157]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.2352	[-0.309, -0.162]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	7.355×10^{-4}	$[-5.89 \times 10^{-4}, 0.00206]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	2.724×10^{-4}	$[-5.41 \times 10^{-4}, 0.00109]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	4.248×10^{-4}	$[-1.61 \times 10^{-4}, 0.00101]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	4.069×10^{-4}	$[-2.00 \times 10^{-4}, 0.00101]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	2.124×10^{-4}	$[-2.27 \times 10^{-4}, 6.52 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	1.519×10^{-4}	$[-6.50 \times 10^{-4}, 9.54 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	5.207×10^{-4}	$[-6.79 \times 10^{-5}, 0.00111]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	4.193×10^{-4}	$[4.60 \times 10^{-5}, 7.92 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.001286	[-0.00419, 0.00162]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.001682	$[-0.00409, 7.24 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.001815	$[-0.00428, 6.52 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.002097	$[-0.00504, 8.49 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.001877	$[-0.00445, 6.92 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00193	$[-0.00395, 8.52 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00183	[-0.00115, 0.00481]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.002918	$[-8.62 \times 10^{-4}, 0.0067]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.06204	[0.0594, 0.0647]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.06256	[0.0605, 0.0646]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.06284	[0.0607, 0.065]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.06272	[0.0605, 0.065]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.06265	[0.0608, 0.0645]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.06364	[0.0599, 0.0673]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.06835	[0.0657, 0.071]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.07828	[0.0734, 0.0831]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1806	[0.144, 0.218]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1875	[0.157, 0.219]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1841	[0.155, 0.213]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1839	[0.153, 0.215]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1793	[0.15, 0.209]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.1924	[0.15, 0.234]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2138	[0.157, 0.271]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.2567	[0.171, 0.343]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.06002	[0.0564, 0.0637]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.0606	[0.0556, 0.0656]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0606	[0.0561, 0.0651]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.06022	[0.0549, 0.0656]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.06056	[0.0561, 0.0651]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.06156	[0.0559, 0.0672]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.06966	[0.066, 0.0733]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.08078	[0.0767, 0.0848]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9886	[0.987, 0.991]	0.00272	5	0
adaptive	pointer	0.01	0.9888	[0.988, 0.99]	0.00163	5	0
adaptive	pointer	0.1	0.989	[0.988, 0.99]	9.94×10^{-4}	5	0
adaptive	pointer	1	0.989	[0.988, 0.99]	9.91×10^{-4}	5	0
adaptive	pointer	10	0.9889	[0.988, 0.99]	0.00111	5	0
adaptive	pointer	100	0.9888	[0.988, 0.99]	0.00119	5	0
adaptive	pointer	1000	0.9894	[0.989, 0.99]	0.00104	5	0
adaptive	pointer	1e+04	0.9896	[0.989, 0.99]	6.07×10^{-4}	5	0
adaptive	seq2seq	0	0.8549	[0.852, 0.857]	0.00321	5	0
adaptive	seq2seq	0.01	0.8538	[0.852, 0.855]	0.00203	5	0
adaptive	seq2seq	0.1	0.8537	[0.851, 0.855]	0.00275	5	0
adaptive	seq2seq	1	0.8539	[0.852, 0.856]	0.00236	5	0
adaptive	seq2seq	10	0.8539	[0.852, 0.855]	0.00224	5	0
adaptive	seq2seq	100	0.8528	[0.849, 0.857]	0.00533	5	0
adaptive	seq2seq	1000	0.8469	[0.844, 0.849]	0.00328	5	0
adaptive	seq2seq	1e+04	0.8191	[0.81, 0.829]	0.0121	5	0
tropical	pointer	0	0.8413	[0.791, 0.888]	0.0634	5	0
tropical	pointer	0.01	0.8467	[0.802, 0.888]	0.0579	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.849	[0.798, 0.894]	0.0623	5	0
tropical	pointer	1	0.8421	[0.794, 0.886]	0.0572	5	0
tropical	pointer	10	0.8457	[0.797, 0.884]	0.0561	5	0
tropical	pointer	100	0.8448	[0.799, 0.887]	0.0587	5	0
tropical	pointer	1000	0.85	[0.81, 0.893]	0.0541	5	0
tropical	pointer	1e+04	0.8465	[0.8, 0.889]	0.0562	5	0
tropical	seq2seq	0	0.6171	[0.604, 0.628]	0.0158	5	0
tropical	seq2seq	0.01	0.6065	[0.588, 0.621]	0.0222	5	0
tropical	seq2seq	0.1	0.602	[0.586, 0.617]	0.0209	5	0
tropical	seq2seq	1	0.6171	[0.604, 0.63]	0.0175	5	0
tropical	seq2seq	10	0.6121	[0.587, 0.63]	0.0283	5	0
tropical	seq2seq	100	0.5948	[0.565, 0.621]	0.0362	5	0
tropical	seq2seq	1000	0.5683	[0.519, 0.616]	0.0621	5	0
tropical	seq2seq	1e+04	0.5059	[0.433, 0.576]	0.0939	5	0
vanilla	pointer	0	0.9867	[0.986, 0.988]	0.00116	5	0
vanilla	pointer	0.01	0.988	[0.987, 0.989]	0.00134	5	0
vanilla	pointer	0.1	0.9878	[0.987, 0.989]	0.00106	5	0
vanilla	pointer	1	0.988	[0.987, 0.989]	0.0012	5	0
vanilla	pointer	10	0.9883	[0.988, 0.989]	0.00115	5	0
vanilla	pointer	100	0.9883	[0.988, 0.989]	0.0011	5	0
vanilla	pointer	1000	0.988	[0.987, 0.989]	0.0011	5	0
vanilla	pointer	1e+04	0.9882	[0.988, 0.989]	6.96×10^{-4}	5	0
vanilla	seq2seq	0	0.8558	[0.851, 0.863]	0.00769	5	0
vanilla	seq2seq	0.01	0.8572	[0.851, 0.864]	0.00829	5	0
vanilla	seq2seq	0.1	0.8567	[0.851, 0.863]	0.0083	5	0
vanilla	seq2seq	1	0.8575	[0.852, 0.864]	0.00854	5	0
vanilla	seq2seq	10	0.8574	[0.853, 0.863]	0.00735	5	0
vanilla	seq2seq	100	0.8553	[0.85, 0.861]	0.00705	5	0
vanilla	seq2seq	1000	0.8429	[0.838, 0.848]	0.00659	5	0
vanilla	seq2seq	1e+04	0.8126	[0.806, 0.819]	0.00862	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	1.400×10^{-4}	[-0.00234, 0.00262]	1.0000	
adaptive	pointer	0.1	3.200×10^{-4}	[-0.00216, 0.0028]	0.9994	
adaptive	pointer	1	3.200×10^{-4}	[-0.00216, 0.0028]	0.9994	
adaptive	pointer	10	3.000×10^{-4}	[-0.00218, 0.00278]	0.9996	
adaptive	pointer	100	1.400×10^{-4}	[-0.00234, 0.00262]	1.0000	
adaptive	pointer	1000	7.400×10^{-4}	[-0.00174, 0.00322]	0.9351	
adaptive	pointer	1e+04	9.200×10^{-4}	[-0.00156, 0.0034]	0.8410	
adaptive	seq2seq	0.01	-0.00112	[-0.0103, 0.00801]	0.9996	
adaptive	seq2seq	0.1	-0.0012	[-0.0103, 0.00793]	0.9994	
adaptive	seq2seq	1	-0.00102	[-0.0102, 0.00811]	0.9998	
adaptive	seq2seq	10	-0.00102	[-0.0102, 0.00811]	0.9998	
adaptive	seq2seq	100	-0.00206	[-0.0112, 0.00707]	0.9839	
adaptive	seq2seq	1000	-0.00802	[-0.0172, 0.00111]	0.1042	
adaptive	seq2seq	1e+04	-0.03582	[-0.045, -0.0267]	$< 10^{-4}$	*
tropical	pointer	0.01	0.00542	[-0.0964, 0.107]	1.0000	
tropical	pointer	0.1	0.00764	[-0.0942, 0.11]	1.0000	
tropical	pointer	1	7.400×10^{-4}	[-0.101, 0.103]	1.0000	
tropical	pointer	10	0.00436	[-0.0975, 0.106]	1.0000	
tropical	pointer	100	0.00348	[-0.0984, 0.105]	1.0000	

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
tropical	pointer	1000	0.00872	[-0.0931, 0.111]	1.0000	
tropical	pointer	1e+04	0.00516	[-0.0967, 0.107]	1.0000	
tropical	seq2seq	0.01	-0.0106	[-0.0894, 0.0682]	0.9993	
tropical	seq2seq	0.1	-0.01512	[-0.0939, 0.0637]	0.9935	
tropical	seq2seq	1	-4.000×10^{-5}	[-0.0788, 0.0787]	1.0000	
tropical	seq2seq	10	-0.00498	[-0.0838, 0.0738]	1.0000	
tropical	seq2seq	100	-0.02232	[-0.101, 0.0565]	0.9488	
tropical	seq2seq	1000	-0.04884	[-0.128, 0.0299]	0.3826	
tropical	seq2seq	1e+04	-0.1112	[-0.19, -0.0324]	0.0029	*
vanilla	pointer	0.01	0.00134	$[-6.03 \times 10^{-4}, 0.00328]$	0.2804	
vanilla	pointer	0.1	0.00114	$[-8.03 \times 10^{-4}, 0.00308]$	0.4389	
vanilla	pointer	1	0.00134	$[-6.03 \times 10^{-4}, 0.00328]$	0.2804	
vanilla	pointer	10	0.00156	$[-3.83 \times 10^{-4}, 0.0035]$	0.1585	
vanilla	pointer	100	0.00164	$[-3.03 \times 10^{-4}, 0.00358]$	0.1266	
vanilla	pointer	1000	0.00128	$[-6.63 \times 10^{-4}, 0.00322]$	0.3232	
vanilla	pointer	1e+04	0.0015	$[-4.43 \times 10^{-4}, 0.00344]$	0.1864	
vanilla	seq2seq	0.01	0.00138	[-0.0123, 0.0151]	0.9999	
vanilla	seq2seq	0.1	9.000×10^{-4}	[-0.0128, 0.0146]	1.0000	
vanilla	seq2seq	1	0.00172	[-0.012, 0.0154]	0.9995	
vanilla	seq2seq	10	0.00158	[-0.0121, 0.0153]	0.9997	
vanilla	seq2seq	100	-5.000×10^{-4}	[-0.0142, 0.0132]	1.0000	
vanilla	seq2seq	1000	-0.0129	$[-0.0266, 7.87 \times 10^{-4}]$	0.0712	
vanilla	seq2seq	1e+04	-0.04322	[-0.0569, -0.0295]	$< 10^{-4}$	*

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	1.071×10^{-4}	$[-7.86 \times 10^{-6}, 2.22 \times 10^{-4}]$	0.2337	0.135	0.3139
adaptive	seq2seq	-0.004242	[-0.00523, -0.00307]	$< 10^{-4}$	-0.501	0.0004
tropical	pointer	1.471×10^{-4}	$[-9.05 \times 10^{-4}, 8.58 \times 10^{-4}]$	0.9740	0.00571	0.9758
tropical	seq2seq	-0.01398	[-0.0235, -0.00515]	0.0022	-0.299	0.0258
vanilla	pointer	3.786×10^{-5}	$[-2.86 \times 10^{-5}, 1.34 \times 10^{-4}]$	0.6663	0.0781	0.5603
vanilla	seq2seq	-0.005844	[-0.00645, -0.0052]	$< 10^{-4}$	-0.528	0.0004

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.1454	[-0.225, -0.0659]	0.0401	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.1413	[-0.215, -0.0679]	0.0401	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.1389	[-0.218, -0.0603]	0.0401	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.146	[-0.218, -0.0735]	0.0401	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.1426	[-0.214, -0.0716]	0.0401	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.1435	[-0.218, -0.0693]	0.0401	*

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.1379	[-0.206, -0.0695]	0.0401	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.1417	[-0.212, -0.0711]	0.0401	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.2387	[-0.267, -0.211]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.2506	[-0.287, -0.215]	0.0005	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.2547	[-0.287, -0.222]	0.0004	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.2404	[-0.268, -0.213]	0.0003	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.2452	[-0.289, -0.201]	0.0011	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.2605	[-0.306, -0.215]	0.0011	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.2746	[-0.353, -0.197]	0.0062	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.3067	[-0.424, -0.189]	0.0172	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00194	[-0.00103, 0.00491]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	7.400×10^{-4}	[-0.00139, 0.00287]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00112	$[-2.61 \times 10^{-4}, 0.0025]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	9.200×10^{-4}	$[-3.51 \times 10^{-4}, 0.00219]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	6.800×10^{-4}	$[-4.22 \times 10^{-4}, 0.00178]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	4.400×10^{-4}	[-0.00149, 0.00237]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0014	$[-1.97 \times 10^{-4}, 0.003]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00136	$[4.71 \times 10^{-4}, 0.00225]$	0.2112	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-9.000×10^{-4}	[-0.00876, 0.00696]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.0034	[-0.0113, 0.00453]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.003	[-0.0108, 0.00477]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00364	[-0.0118, 0.00456]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0035	[-0.0108, 0.00375]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00246	[-0.00701, 0.00209]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00398	[-0.00243, 0.0104]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0065	[-0.00284, 0.0158]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1338	[0.127, 0.141]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.135	[0.133, 0.137]	$< 10^{-4}$	*

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1353	[0.132, 0.138]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1351	[0.132, 0.138]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1351	[0.133, 0.138]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.136	[0.13, 0.142]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.1425	[0.139, 0.146]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1705	[0.156, 0.185]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.2242	[0.131, 0.318]	0.0128	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.2402	[0.152, 0.328]	0.0121	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.247	[0.158, 0.336]	0.0121	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.225	[0.135, 0.315]	0.0128	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2335	[0.14, 0.327]	0.0128	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.25	[0.152, 0.348]	0.0128	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2818	[0.153, 0.411]	0.0128	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.3406	[0.18, 0.501]	0.0128	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1309	[0.121, 0.141]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1309	[0.12, 0.142]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1312	[0.12, 0.142]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1305	[0.119, 0.142]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1309	[0.121, 0.141]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1331	[0.123, 0.143]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1451	[0.136, 0.154]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1756	[0.164, 0.187]	$< 10^{-4}$	*

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.4113	[0.4, 0.422]	0.0142	5	0
adaptive	pointer	0.01	0.4209	[0.406, 0.433]	0.0178	5	0
adaptive	pointer	0.1	0.4198	[0.406, 0.431]	0.0165	5	0
adaptive	pointer	1	0.4204	[0.407, 0.431]	0.0163	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.4207	[0.406, 0.431]	0.0161	5	0
adaptive	pointer	100	0.4212	[0.408, 0.431]	0.0159	5	0
adaptive	pointer	1000	0.4247	[0.411, 0.436]	0.0165	5	0
adaptive	pointer	1e+04	0.4272	[0.411, 0.439]	0.0181	5	0
adaptive	seq2seq	0	0.4298	[0.422, 0.439]	0.0117	5	0
adaptive	seq2seq	0.01	0.4472	[0.434, 0.465]	0.0206	5	0
adaptive	seq2seq	0.1	0.4466	[0.433, 0.464]	0.0201	5	0
adaptive	seq2seq	1	0.4457	[0.433, 0.463]	0.0189	5	0
adaptive	seq2seq	10	0.4466	[0.434, 0.466]	0.0214	5	0
adaptive	seq2seq	100	0.4461	[0.434, 0.464]	0.0184	5	0
adaptive	seq2seq	1000	0.4438	[0.43, 0.463]	0.022	5	0
adaptive	seq2seq	1e+04	0.4342	[0.421, 0.451]	0.0205	5	0
tropical	pointer	0	0.5436	[0.476, 0.612]	0.0859	5	0
tropical	pointer	0.01	0.5411	[0.486, 0.596]	0.0716	5	0
tropical	pointer	0.1	0.5264	[0.467, 0.588]	0.0782	5	0
tropical	pointer	1	0.5272	[0.459, 0.596]	0.0896	5	0
tropical	pointer	10	0.5463	[0.496, 0.594]	0.0634	5	0
tropical	pointer	100	0.543	[0.459, 0.606]	0.0938	5	0
tropical	pointer	1000	0.537	[0.468, 0.606]	0.0875	5	0
tropical	pointer	1e+04	0.5398	[0.458, 0.599]	0.0912	5	0
tropical	seq2seq	0	0.4613	[0.427, 0.497]	0.0464	5	0
tropical	seq2seq	0.01	0.4845	[0.436, 0.535]	0.0607	5	0
tropical	seq2seq	0.1	0.4809	[0.439, 0.524]	0.0538	5	0
tropical	seq2seq	1	0.4888	[0.452, 0.526]	0.0474	5	0
tropical	seq2seq	10	0.4718	[0.426, 0.527]	0.0605	5	0
tropical	seq2seq	100	0.4743	[0.427, 0.531]	0.0704	5	0
tropical	seq2seq	1000	0.4901	[0.431, 0.549]	0.0767	5	0
tropical	seq2seq	1e+04	0.4802	[0.392, 0.568]	0.113	5	0
vanilla	pointer	0	0.4056	[0.394, 0.414]	0.0127	5	0
vanilla	pointer	0.01	0.4225	[0.407, 0.435]	0.0176	5	0
vanilla	pointer	0.1	0.4215	[0.408, 0.433]	0.0165	5	0
vanilla	pointer	1	0.423	[0.409, 0.435]	0.0167	5	0
vanilla	pointer	10	0.4217	[0.408, 0.433]	0.016	5	0
vanilla	pointer	100	0.4229	[0.408, 0.434]	0.017	5	0
vanilla	pointer	1000	0.4207	[0.408, 0.432]	0.0168	5	0
vanilla	pointer	1e+04	0.4192	[0.406, 0.431]	0.0171	5	0
vanilla	seq2seq	0	0.4322	[0.422, 0.448]	0.0166	5	0
vanilla	seq2seq	0.01	0.4409	[0.426, 0.459]	0.0216	5	0
vanilla	seq2seq	0.1	0.4412	[0.425, 0.458]	0.0206	5	0
vanilla	seq2seq	1	0.4407	[0.427, 0.453]	0.0173	5	0
vanilla	seq2seq	10	0.4417	[0.43, 0.453]	0.0153	5	0
vanilla	seq2seq	100	0.4482	[0.429, 0.47]	0.0257	5	0
vanilla	seq2seq	1000	0.4482	[0.431, 0.468]	0.0254	5	0
vanilla	seq2seq	1e+04	0.4338	[0.423, 0.45]	0.018	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.009602	[-0.0192, 0.0384]	0.8945
adaptive	pointer	0.1	0.008526	[-0.0202, 0.0373]	0.9370
adaptive	pointer	1	0.009082	[-0.0197, 0.0379]	0.9166
adaptive	pointer	10	0.009397	[-0.0194, 0.0382]	0.9036
adaptive	pointer	100	0.009906	[-0.0189, 0.0387]	0.8803

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	1000	0.0134	[-0.0154, 0.0422]	0.6685
adaptive	pointer	1e+04	0.01594	[-0.0128, 0.0447]	0.4981
adaptive	seq2seq	0.01	0.01738	[-0.0166, 0.0513]	0.5785
adaptive	seq2seq	0.1	0.01682	[-0.0171, 0.0508]	0.6107
adaptive	seq2seq	1	0.01594	[-0.018, 0.0499]	0.6611
adaptive	seq2seq	10	0.01679	[-0.0172, 0.0508]	0.6122
adaptive	seq2seq	100	0.0163	[-0.0177, 0.0503]	0.6403
adaptive	seq2seq	1000	0.01407	[-0.0199, 0.048]	0.7666
adaptive	seq2seq	1e+04	0.004418	[-0.0295, 0.0384]	0.9994
tropical	pointer	0.01	-0.002559	[-0.148, 0.143]	1.0000
tropical	pointer	0.1	-0.01728	[-0.163, 0.128]	0.9997
tropical	pointer	1	-0.01642	[-0.162, 0.129]	0.9998
tropical	pointer	10	0.002618	[-0.143, 0.148]	1.0000
tropical	pointer	100	-6.931×10^{-4}	[-0.146, 0.145]	1.0000
tropical	pointer	1000	-0.006681	[-0.152, 0.139]	1.0000
tropical	pointer	1e+04	-0.003891	[-0.149, 0.142]	1.0000
tropical	seq2seq	0.01	0.02323	[-0.0975, 0.144]	0.9934
tropical	seq2seq	0.1	0.01958	[-0.101, 0.14]	0.9976
tropical	seq2seq	1	0.02748	[-0.0933, 0.148]	0.9831
tropical	seq2seq	10	0.01054	[-0.11, 0.131]	1.0000
tropical	seq2seq	100	0.01297	[-0.108, 0.134]	0.9998
tropical	seq2seq	1000	0.02886	[-0.0919, 0.15]	0.9781
tropical	seq2seq	1e+04	0.01887	[-0.102, 0.14]	0.9981
vanilla	pointer	0.01	0.01697	[-0.0116, 0.0455]	0.4260
vanilla	pointer	0.1	0.01593	[-0.0126, 0.0445]	0.4906
vanilla	pointer	1	0.01745	[-0.0111, 0.046]	0.3975
vanilla	pointer	10	0.01616	[-0.0124, 0.0447]	0.4763
vanilla	pointer	100	0.0173	[-0.0113, 0.0459]	0.4060
vanilla	pointer	1000	0.01515	[-0.0134, 0.0437]	0.5424
vanilla	pointer	1e+04	0.01365	[-0.0149, 0.0422]	0.6446
vanilla	seq2seq	0.01	0.00869	[-0.027, 0.0443]	0.9757
vanilla	seq2seq	0.1	0.008957	[-0.0267, 0.0446]	0.9716
vanilla	seq2seq	1	0.008433	[-0.0272, 0.0441]	0.9792
vanilla	seq2seq	10	0.009445	[-0.0262, 0.0451]	0.9629
vanilla	seq2seq	100	0.01595	[-0.0197, 0.0516]	0.7035
vanilla	seq2seq	1000	0.01602	[-0.0196, 0.0517]	0.6999
vanilla	seq2seq	1e+04	0.001594	[-0.034, 0.0372]	1.0000

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.001056	$[5.20 \times 10^{-4}$, 0.00159]	0.4265	0.23	0.0832
adaptive	seq2seq	-0.001572	[-0.00237, -6.92×10^{-4}]	0.3331	-0.181	0.1796
tropical	pointer	0.001176	[-0.00454, 0.00677]	0.8570	0.0362	0.8072
tropical	seq2seq	-3.221×10^{-4}	[-0.0125, 0.0109]	0.9546	-0.0286	0.8454
vanilla	pointer	-4.170×10^{-4}	[-0.00144, 4.81×10^{-4}]	0.7525	-0.0667	0.6267
vanilla	seq2seq	1.307×10^{-5}	[-0.00219, 0.00206]	0.9938	-0.0438	0.7441

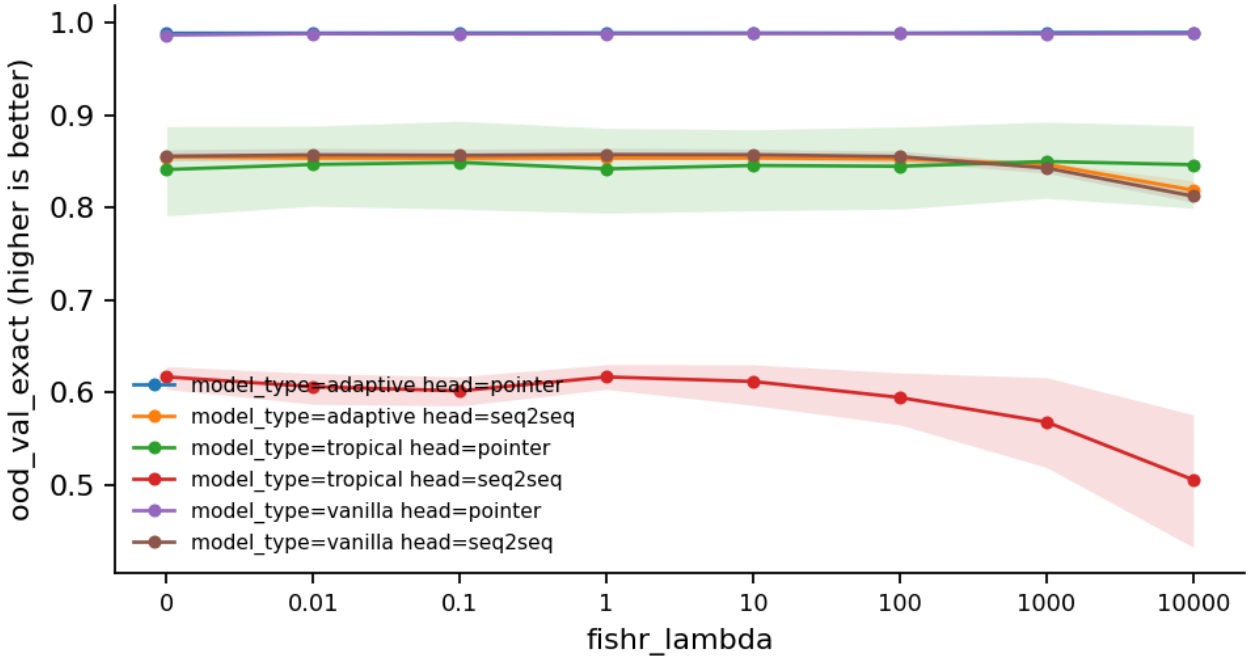


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

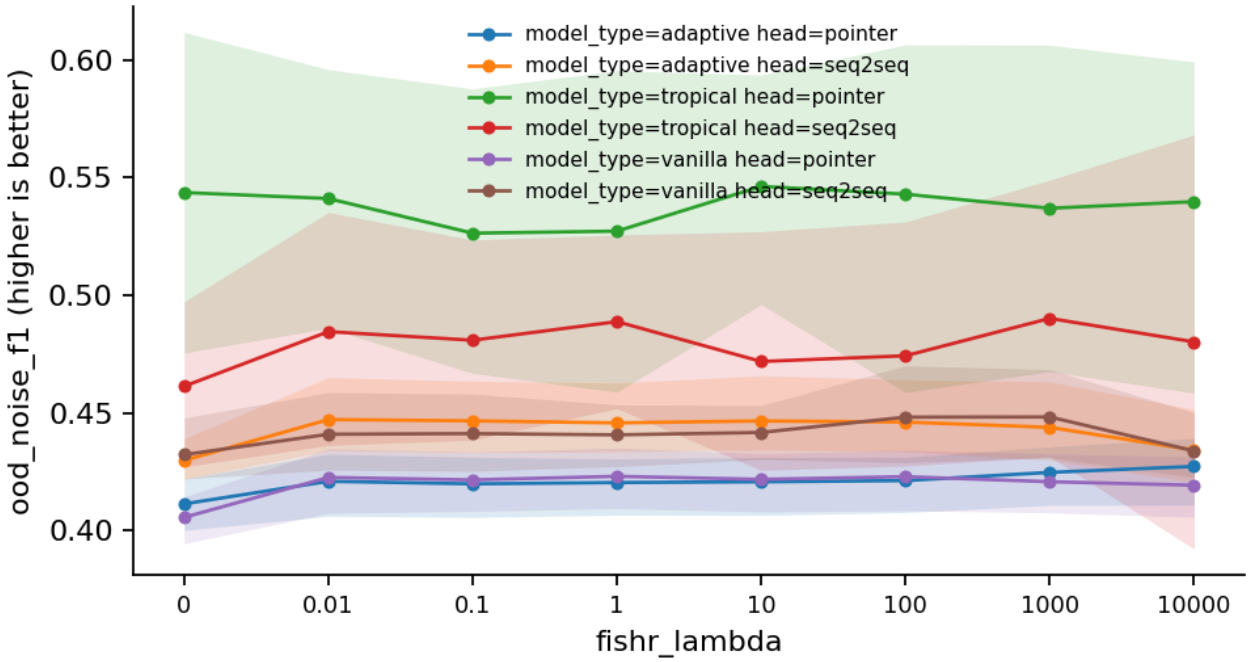


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1381	[0.0177, 0.258]	0.5007
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.1185	[0.0143, 0.223]	0.5007
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1049	[-0.00719, 0.217]	0.6342
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1042	[-0.0216, 0.23]	0.7460
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1245	[0.0332, 0.216]	0.3093
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1201	[-0.00238, 0.243]	0.6342
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1162	[-0.00345, 0.236]	0.6342
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.1205	[0.00372, 0.237]	0.5942
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.02906	[-0.0397, 0.0978]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.0436	[-0.0401, 0.127]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.03968	[-0.0327, 0.112]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.04811	[-0.0187, 0.115]	0.9310
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.03015	[-0.0575, 0.118]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.02608	[-0.052, 0.104]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.0419	[-0.0441, 0.128]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.04634	[-0.0835, 0.176]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.005718	[-0.00405, 0.0155]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.001648	[-0.00827, 0.00497]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.00169	[-0.00938, 0.006]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.002649	[-0.0106, 0.00533]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.001041	[-0.00609, 0.004]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.001677	[-0.0121, 0.00876]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.003972	[-0.00956, 0.0175]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.008003	[-0.00307, 0.0191]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.002449	[-0.0235, 0.0186]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.006237	[-0.0156, 0.028]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.005413	[-0.0131, 0.0239]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.005062	[-0.0162, 0.0263]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.004896	[-0.0205, 0.0302]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.002101	[-0.0234, 0.0192]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.004406	[-0.0203, 0.0115]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	3.747×10^{-4}	[-0.0125, 0.0132]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.01848	[-0.0409, 0.00398]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.02626	[-0.0476, -0.0049]	0.6176
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.02678	[-0.0482, -0.00531]	0.6176
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.02534	[-0.0464, -0.00431]	0.6176
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.02587	[-0.0527, 9.86×10^{-4}]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02488	[-0.045, -0.00474]	0.6176
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01915	[-0.0435, 0.00521]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.006965	[-0.0301, 0.0162]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.08236	[-0.0542, 0.219]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.05657	[-0.0923, 0.205]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.0455	[-0.102, 0.193]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.03846	[-0.123, 0.2]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.07444	[-0.0498, 0.199]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.06869	[-0.122, 0.259]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.04682	[-0.149, 0.243]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.05959	[-0.171, 0.291]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.02665	[-0.0482, -0.00509]	0.6176
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.01837	[-0.0483, 0.0116]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.01967	[-0.0498, 0.0104]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.01763	[-0.0467, 0.0114]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.01994	[-0.0427, 0.0028]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.0253	[-0.064, 0.0134]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.02753	[-0.0675, 0.0124]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.01459	[-0.0439, 0.0147]	1.0000

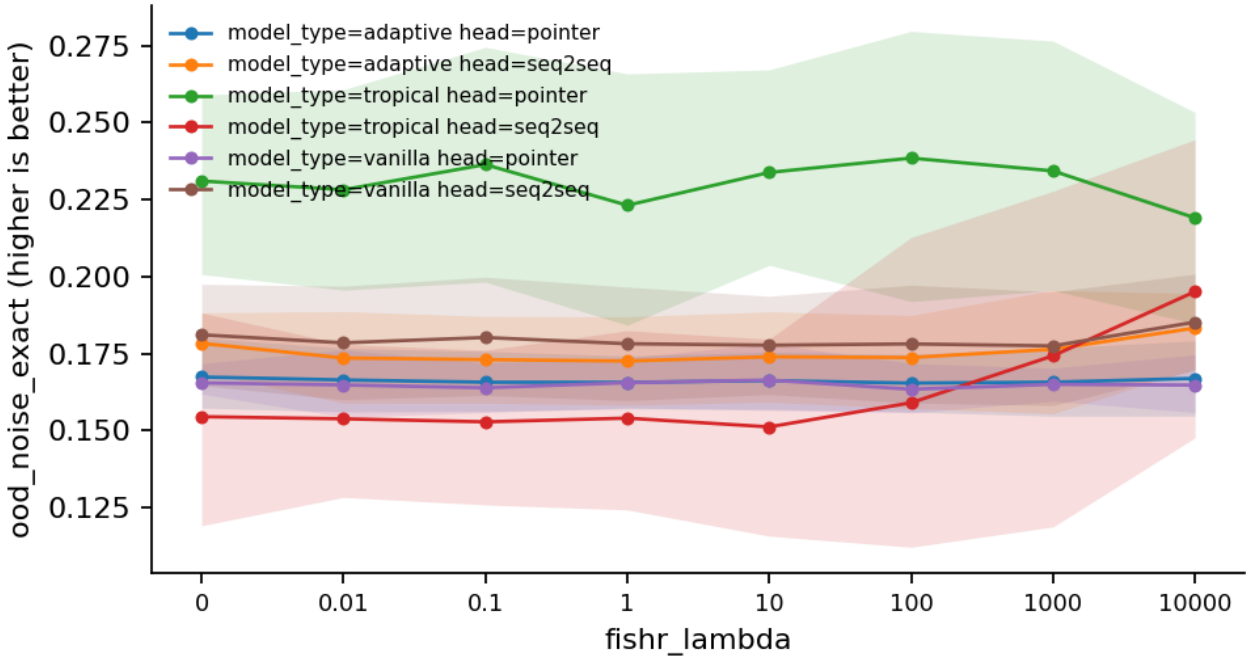


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.1674	[0.157, 0.18]	0.0144	5	0
adaptive	pointer	0.01	0.1665	[0.156, 0.177]	0.0131	5	0
adaptive	pointer	0.1	0.1657	[0.156, 0.176]	0.0127	5	0
adaptive	pointer	1	0.1657	[0.157, 0.174]	0.0111	5	0
adaptive	pointer	10	0.1663	[0.157, 0.176]	0.0119	5	0
adaptive	pointer	100	0.1655	[0.156, 0.175]	0.0118	5	0
adaptive	pointer	1000	0.1658	[0.155, 0.177]	0.0141	5	0
adaptive	pointer	1e+04	0.167	[0.155, 0.179]	0.0153	5	0
adaptive	seq2seq	0	0.1784	[0.169, 0.188]	0.0126	5	0
adaptive	seq2seq	0.01	0.1736	[0.159, 0.189]	0.0196	5	0
adaptive	seq2seq	0.1	0.1731	[0.159, 0.187]	0.0194	5	0
adaptive	seq2seq	1	0.1726	[0.158, 0.187]	0.0195	5	0
adaptive	seq2seq	10	0.174	[0.159, 0.189]	0.0202	5	0
adaptive	seq2seq	100	0.1738	[0.157, 0.187]	0.0196	5	0
adaptive	seq2seq	1000	0.1764	[0.156, 0.196]	0.0255	5	0
adaptive	seq2seq	1e+04	0.1833	[0.17, 0.195]	0.0157	5	0
tropical	pointer	0	0.2311	[0.201, 0.259]	0.0364	5	0
tropical	pointer	0.01	0.2282	[0.196, 0.261]	0.0453	5	0
tropical	pointer	0.1	0.2364	[0.198, 0.275]	0.0492	5	0
tropical	pointer	1	0.2232	[0.184, 0.266]	0.0547	5	0
tropical	pointer	10	0.2338	[0.204, 0.267]	0.0405	5	0
tropical	pointer	100	0.2385	[0.192, 0.28]	0.0542	5	0
tropical	pointer	1000	0.2343	[0.196, 0.276]	0.0521	5	0
tropical	pointer	1e+04	0.2191	[0.185, 0.253]	0.0455	5	0
tropical	seq2seq	0	0.1546	[0.119, 0.188]	0.0427	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.1539	[0.128, 0.178]	0.0325	5	0
tropical	seq2seq	0.1	0.1529	[0.126, 0.176]	0.0317	5	0
tropical	seq2seq	1	0.1541	[0.124, 0.182]	0.0373	5	0
tropical	seq2seq	10	0.1512	[0.116, 0.18]	0.0399	5	0
tropical	seq2seq	100	0.1591	[0.112, 0.213]	0.0674	5	0
tropical	seq2seq	1000	0.1743	[0.119, 0.228]	0.0725	5	0
tropical	seq2seq	1e+04	0.1951	[0.148, 0.245]	0.0622	5	0
vanilla	pointer	0	0.1655	[0.162, 0.172]	0.00691	5	0
vanilla	pointer	0.01	0.1649	[0.154, 0.176]	0.0139	5	0
vanilla	pointer	0.1	0.1639	[0.156, 0.172]	0.0106	5	0
vanilla	pointer	1	0.1656	[0.157, 0.174]	0.0105	5	0
vanilla	pointer	10	0.1664	[0.157, 0.178]	0.0142	5	0
vanilla	pointer	100	0.1635	[0.156, 0.172]	0.0109	5	0
vanilla	pointer	1000	0.165	[0.16, 0.17]	0.00688	5	0
vanilla	pointer	1e+04	0.1648	[0.156, 0.175]	0.0116	5	0
vanilla	seq2seq	0	0.1811	[0.165, 0.198]	0.0207	5	0
vanilla	seq2seq	0.01	0.1785	[0.16, 0.197]	0.0238	5	0
vanilla	seq2seq	0.1	0.1803	[0.162, 0.2]	0.0254	5	0
vanilla	seq2seq	1	0.1782	[0.16, 0.197]	0.0243	5	0
vanilla	seq2seq	10	0.1778	[0.162, 0.194]	0.0213	5	0
vanilla	seq2seq	100	0.1782	[0.159, 0.197]	0.0253	5	0
vanilla	seq2seq	1000	0.1775	[0.158, 0.195]	0.0237	5	0
vanilla	seq2seq	1e+04	0.1853	[0.17, 0.201]	0.0214	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-9.600×10^{-4}	[-0.0239, 0.022]	1.0000
adaptive	pointer	0.1	-0.0017	[-0.0246, 0.0212]	1.0000
adaptive	pointer	1	-0.00174	[-0.0247, 0.0212]	1.0000
adaptive	pointer	10	-0.00118	[-0.0241, 0.0217]	1.0000
adaptive	pointer	100	-0.00198	[-0.0249, 0.0209]	1.0000
adaptive	pointer	1000	-0.00168	[-0.0246, 0.0212]	1.0000
adaptive	pointer	1e+04	-4.600×10^{-4}	[-0.0234, 0.0225]	1.0000
adaptive	seq2seq	0.01	-0.00478	[-0.0386, 0.029]	0.9990
adaptive	seq2seq	0.1	-0.0053	[-0.0391, 0.0285]	0.9981
adaptive	seq2seq	1	-0.00576	[-0.0396, 0.028]	0.9968
adaptive	seq2seq	10	-0.00438	[-0.0382, 0.0294]	0.9994
adaptive	seq2seq	100	-0.00462	[-0.0384, 0.0292]	0.9992
adaptive	seq2seq	1000	-0.00194	[-0.0357, 0.0319]	1.0000
adaptive	seq2seq	1e+04	0.00494	[-0.0289, 0.0387]	0.9987
tropical	pointer	0.01	-0.00282	[-0.0861, 0.0804]	1.0000
tropical	pointer	0.1	0.00534	[-0.0779, 0.0886]	1.0000
tropical	pointer	1	-0.00788	[-0.0911, 0.0754]	0.9999
tropical	pointer	10	0.00278	[-0.0805, 0.086]	1.0000
tropical	pointer	100	0.00744	[-0.0758, 0.0907]	0.9999
tropical	pointer	1000	0.00328	[-0.08, 0.0865]	1.0000
tropical	pointer	1e+04	-0.01196	[-0.0952, 0.0713]	0.9989
tropical	seq2seq	0.01	-7.000×10^{-4}	[-0.0892, 0.0878]	1.0000
tropical	seq2seq	0.1	-0.00168	[-0.0902, 0.0868]	1.0000
tropical	seq2seq	1	-4.800×10^{-4}	[-0.089, 0.088]	1.0000
tropical	seq2seq	10	-0.00332	[-0.0918, 0.0852]	1.0000
tropical	seq2seq	100	0.00452	[-0.084, 0.093]	1.0000

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
tropical	seq2seq	1000	0.01976	[-0.0688, 0.108]	0.9848
tropical	seq2seq	1e+04	0.04058	[-0.0479, 0.129]	0.6829
vanilla	pointer	0.01	-6.600×10^{-4}	[-0.0198, 0.0185]	1.0000
vanilla	pointer	0.1	-0.00158	[-0.0208, 0.0176]	1.0000
vanilla	pointer	1	6.000×10^{-5}	[-0.0191, 0.0192]	1.0000
vanilla	pointer	10	9.200×10^{-4}	[-0.0183, 0.0201]	1.0000
vanilla	pointer	100	-0.00206	[-0.0212, 0.0171]	0.9998
vanilla	pointer	1000	-4.800×10^{-4}	[-0.0197, 0.0187]	1.0000
vanilla	pointer	1e+04	-7.000×10^{-4}	[-0.0199, 0.0185]	1.0000
vanilla	seq2seq	0.01	-0.0026	[-0.0433, 0.0381]	1.0000
vanilla	seq2seq	0.1	-8.800×10^{-4}	[-0.0416, 0.0398]	1.0000
vanilla	seq2seq	1	-0.0029	[-0.0436, 0.0378]	1.0000
vanilla	seq2seq	10	-0.00338	[-0.0441, 0.0373]	1.0000
vanilla	seq2seq	100	-0.00296	[-0.0437, 0.0377]	1.0000
vanilla	seq2seq	1000	-0.0036	[-0.0443, 0.0371]	1.0000
vanilla	seq2seq	1e+04	0.00412	[-0.0366, 0.0448]	0.9999

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	4.643×10^{-5}	$[-2.82 \times 10^{-4}, 3.69 \times 10^{-4}]$	0.9635	0	1.0000
adaptive	seq2seq	0.001322	$[4.59 \times 10^{-4}, 0.00253]$	0.4085	0.131	0.3291
tropical	pointer	-5.793×10^{-4}	[-0.00395, 0.00188]	0.8817	-0.04	0.7730
tropical	seq2seq	0.006133	[-0.00215, 0.017]	0.1439	0.139	0.3021
vanilla	pointer	-1.429×10^{-6}	$[-9.56 \times 10^{-4}, 7.86 \times 10^{-4}]$	0.9987	0.0171	0.9006
vanilla	seq2seq	5.236×10^{-4}	$[-9.02 \times 10^{-4}, 0.00231]$	0.7792	0.0152	0.9164

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.06554	[0.0198, 0.111]	0.2622
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06338	[-0.00394, 0.131]	0.7099
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.07246	[0.00282, 0.142]	0.6692
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.0576	[-0.0202, 0.135]	0.9810
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.0674	[0.00183, 0.133]	0.6692
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.07504	[-0.00514, 0.155]	0.7099
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0693	[0.00176, 0.137]	0.6692
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.05428	[-0.0129, 0.121]	0.8826
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.02658	[-0.0942, 0.0411]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.02468	[-0.0742, 0.0248]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.02738	[-0.0719, 0.0172]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.02416	[-0.0752, 0.0269]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.02652	[-0.073, 0.0199]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.0191	[-0.0854, 0.0472]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.00322	[-0.0762, 0.0698]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.00988	[-0.0497, 0.0695]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00192	[-0.0219, 0.0258]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00162	[-0.00233, 0.00557]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.0018	[-0.00252, 0.00612]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	1.200×10^{-4}	[-0.00257, 0.00281]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-1.800×10^{-4}	[-0.00505, 0.00469]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.002	[-0.0061, 0.0101]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	7.200×10^{-4}	[-0.011, 0.0124]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00216	[-0.00485, 0.00917]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.00276	[-0.0174, 0.0119]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00494	[-0.013, 0.00316]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00718	[-0.017, 0.00264]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.00562	[-0.0135, 0.00223]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.00376	[-0.00989, 0.00237]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00442	[-0.018, 0.0092]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.0011	[-0.00834, 0.00614]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.00194	[-0.0101, 0.00622]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.01094	[-0.0412, 0.0193]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.00712	[-0.0283, 0.0141]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.00734	[-0.0287, 0.014]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.00692	[-0.0271, 0.0132]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.00774	[-0.0296, 0.0141]	1.0000

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.0083	[-0.028, 0.0114]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01068	[-0.0393, 0.0179]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.01634	[-0.0351, 0.00242]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0765	[-0.00548, 0.158]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.07438	[-0.0115, 0.16]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.08352	[-0.0083, 0.175]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.0691	[-0.0343, 0.172]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.0826	[-0.00546, 0.171]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.07942	[-0.0689, 0.228]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.06002	[-0.0848, 0.205]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.02396	[-0.0947, 0.143]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.01562	[-0.0374, 0.0062]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.01368	[-0.0429, 0.0155]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.01632	[-0.0465, 0.0138]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.01266	[-0.0408, 0.0154]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.01132	[-0.0386, 0.016]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.01472	[-0.0435, 0.0141]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0125	[-0.0359, 0.0109]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.02044	[-0.0413, 4.28×10^{-4}]	1.0000

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9976	[0.997, 0.998]	7.82×10^{-4}	5	0
adaptive	pointer	0.01	0.9977	[0.997, 0.998]	2.46×10^{-4}	5	0
adaptive	pointer	0.1	0.9976	[0.997, 0.998]	3.20×10^{-4}	5	0
adaptive	pointer	1	0.9975	[0.997, 0.998]	2.93×10^{-4}	5	0
adaptive	pointer	10	0.9975	[0.997, 0.998]	3.63×10^{-4}	5	0
adaptive	pointer	100	0.9976	[0.997, 0.998]	3.92×10^{-4}	5	0
adaptive	pointer	1000	0.9977	[0.997, 0.998]	3.74×10^{-4}	5	0
adaptive	pointer	1e+04	0.9977	[0.998, 0.998]	1.58×10^{-4}	5	0
adaptive	seq2seq	0	0.9391	[0.938, 0.941]	0.00206	5	0
adaptive	seq2seq	0.01	0.9389	[0.937, 0.941]	0.00246	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.939	[0.938, 0.94]	0.00188	5	0
adaptive	seq2seq	1	0.9391	[0.937, 0.941]	0.00247	5	0
adaptive	seq2seq	10	0.939	[0.937, 0.941]	0.00219	5	0
adaptive	seq2seq	100	0.9379	[0.935, 0.941]	0.00382	5	0
adaptive	seq2seq	1000	0.934	[0.932, 0.936]	0.0029	5	0
adaptive	seq2seq	1e+04	0.9259	[0.922, 0.931]	0.00557	5	0
tropical	pointer	0	0.943	[0.922, 0.96]	0.0252	5	0
tropical	pointer	0.01	0.9458	[0.929, 0.961]	0.0218	5	0
tropical	pointer	0.1	0.9469	[0.928, 0.963]	0.0232	5	0
tropical	pointer	1	0.9446	[0.927, 0.96]	0.0211	5	0
tropical	pointer	10	0.9452	[0.925, 0.96]	0.0212	5	0
tropical	pointer	100	0.9446	[0.926, 0.961]	0.023	5	0
tropical	pointer	1000	0.947	[0.932, 0.962]	0.0203	5	0
tropical	pointer	1e+04	0.9461	[0.929, 0.961]	0.022	5	0
tropical	seq2seq	0	0.7523	[0.744, 0.76]	0.0103	5	0
tropical	seq2seq	0.01	0.7484	[0.744, 0.754]	0.00611	5	0
tropical	seq2seq	0.1	0.7523	[0.748, 0.757]	0.00566	5	0
tropical	seq2seq	1	0.7493	[0.746, 0.754]	0.00536	5	0
tropical	seq2seq	10	0.7549	[0.751, 0.76]	0.00594	5	0
tropical	seq2seq	100	0.7421	[0.722, 0.755]	0.023	5	0
tropical	seq2seq	1000	0.7231	[0.698, 0.745]	0.0298	5	0
tropical	seq2seq	1e+04	0.6818	[0.633, 0.72]	0.0556	5	0
vanilla	pointer	0	0.9969	[0.997, 0.997]	2.02×10^{-4}	5	0
vanilla	pointer	0.01	0.9974	[0.997, 0.998]	3.76×10^{-4}	5	0
vanilla	pointer	0.1	0.9974	[0.997, 0.998]	3.11×10^{-4}	5	0
vanilla	pointer	1	0.9974	[0.997, 0.998]	2.53×10^{-4}	5	0
vanilla	pointer	10	0.9974	[0.997, 0.998]	3.31×10^{-4}	5	0
vanilla	pointer	100	0.9973	[0.997, 0.998]	4.28×10^{-4}	5	0
vanilla	pointer	1000	0.9974	[0.997, 0.998]	4.58×10^{-4}	5	0
vanilla	pointer	1e+04	0.9976	[0.997, 0.998]	2.98×10^{-4}	5	0
vanilla	seq2seq	0	0.9412	[0.939, 0.944]	0.00353	5	0
vanilla	seq2seq	0.01	0.9414	[0.939, 0.945]	0.00411	5	0
vanilla	seq2seq	0.1	0.9409	[0.938, 0.945]	0.00445	5	0
vanilla	seq2seq	1	0.9417	[0.939, 0.946]	0.00463	5	0
vanilla	seq2seq	10	0.9415	[0.938, 0.945]	0.00441	5	0
vanilla	seq2seq	100	0.9405	[0.937, 0.944]	0.00446	5	0
vanilla	seq2seq	1000	0.9327	[0.93, 0.935]	0.00302	5	0
vanilla	seq2seq	1e+04	0.9227	[0.919, 0.926]	0.00493	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	1.633×10^{-4}	$[-6.27 \times 10^{-4}, 8.00 \times 10^{-4}]$	1.0000
adaptive	pointer	0.1	-1.302×10^{-4}	$[-7.05 \times 10^{-4}, 8.06 \times 10^{-4}]$	1.0000
adaptive	pointer	1	-1.952×10^{-4}	$[-7.38 \times 10^{-4}, 6.63 \times 10^{-4}]$	1.0000
adaptive	pointer	10	6.044×10^{-7}	$[-8.54 \times 10^{-4}, 7.28 \times 10^{-4}]$	1.0000
adaptive	pointer	100	0	$[-8.53 \times 10^{-4}, 7.48 \times 10^{-4}]$	1.0000
adaptive	pointer	1000	3.261×10^{-4}	$[-6.91 \times 10^{-4}, 7.41 \times 10^{-4}]$	1.0000
adaptive	pointer	1e+04	1.628×10^{-4}	$[-5.36 \times 10^{-4}, 7.80 \times 10^{-4}]$	1.0000
adaptive	seq2seq	0.01	-1.046×10^{-4}	$[-7.10 \times 10^{-4}, 2.04 \times 10^{-4}]$	1.0000
adaptive	seq2seq	0.1	1.939×10^{-5}	$[-5.76 \times 10^{-4}, 5.85 \times 10^{-4}]$	1.0000
adaptive	seq2seq	1	3.703×10^{-4}	$[-4.62 \times 10^{-4}, 3.94 \times 10^{-4}]$	1.0000
adaptive	seq2seq	10	1.464×10^{-4}	$[-6.97 \times 10^{-4}, 4.93 \times 10^{-4}]$	1.0000

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	seq2seq	100	-5.759×10^{-4}	[-0.0031, 4.83×10^{-4}]	1.0000
adaptive	seq2seq	1000	-0.005027	[-0.0062, -0.00399]	0.4375
adaptive	seq2seq	1e+04	-0.01462	[-0.0165, -0.00929]	0.4375
tropical	pointer	0.01	0.002587	[-1.18×10^{-4}, 0.00617]	1.0000
tropical	pointer	0.1	0.003549	[0.00105, 0.00672]	0.8750
tropical	pointer	1	5.463×10^{-4}	[-0.00131, 0.00529]	1.0000
tropical	pointer	10	0.002045	[-9.51×10^{-4}, 0.00564]	1.0000
tropical	pointer	100	0.001366	[-2.06×10^{-4}, 0.00359]	1.0000
tropical	pointer	1000	0.002481	[-1.98×10^{-5}, 0.00877]	1.0000
tropical	pointer	1e+04	0.004979	[-0.00243, 0.00735]	1.0000
tropical	seq2seq	0.01	-0.004348	[-0.00747, -1.51×10^{-4}]	0.9375
tropical	seq2seq	0.1	1.289×10^{-4}	[-0.00701, 0.00703]	1.0000
tropical	seq2seq	1	-4.252×10^{-4}	[-0.00756, 0.00139]	1.0000
tropical	seq2seq	10	0.007708	[-0.00605, 0.0113]	1.0000
tropical	seq2seq	100	-0.005418	[-0.0286, 0.0029]	1.0000
tropical	seq2seq	1000	-0.02253	[-0.0502, -0.0114]	0.4375
tropical	seq2seq	1e+04	-0.04	[-0.113, -0.0379]	0.4375
vanilla	pointer	0.01	5.484×10^{-4}	[1.88×10^{-4}, 8.25×10^{-4}]	0.4375
vanilla	pointer	0.1	7.168×10^{-4}	[2.20×10^{-4}, 7.98×10^{-4}]	0.4375
vanilla	pointer	1	5.807×10^{-4}	[2.93×10^{-4}, 7.27×10^{-4}]	0.4375
vanilla	pointer	10	7.818×10^{-4}	[2.35×10^{-4}, 8.80×10^{-4}]	0.4375
vanilla	pointer	100	4.185×10^{-4}	[7.02×10^{-5}, 7.86×10^{-4}]	0.4375
vanilla	pointer	1000	6.451×10^{-4}	[5.73×10^{-5}, 8.96×10^{-4}]	0.4375
vanilla	pointer	1e+04	9.084×10^{-4}	[3.25×10^{-4}, 9.99×10^{-4}]	0.4375
vanilla	seq2seq	0.01	1.891×10^{-4}	[-6.33×10^{-4}, 8.13×10^{-4}]	1.0000
vanilla	seq2seq	0.1	-3.166×10^{-4}	[-0.00102, 5.12×10^{-4}]	1.0000
vanilla	seq2seq	1	8.694×10^{-4}	[-5.09×10^{-4}, 0.0016]	1.0000
vanilla	seq2seq	10	7.210×10^{-4}	[-6.74×10^{-4}, 0.00117]	1.0000
vanilla	seq2seq	100	-8.893×10^{-4}	[-0.00171, 7.02×10^{-4}]	1.0000
vanilla	seq2seq	1000	-0.007102	[-0.0101, -0.00698]	0.4375
vanilla	seq2seq	1e+04	-0.01801	[-0.0208, -0.0159]	0.4375

Table 27: `id_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.09625	[-0.158, 0.281]	0.5823	0.059	0.6693
adaptive	seq2seq	-0.628	[-0.825, -0.503]	$< 10^{-4}$	-0.497	0.0002
tropical	pointer	0.004243	[-0.221, 0.217]	0.9807	0.00952	0.9512
tropical	seq2seq	-0.5106	[-0.634, -0.33]	0.0017	-0.383	0.0042
vanilla	pointer	0.06932	[-0.103, 0.325]	0.6923	0.0514	0.7095
vanilla	seq2seq	-0.6592	[-0.77, -0.529]	$< 10^{-4}$	-0.543	0.0002

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.05389	[-0.0851, -0.0227]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.05162	[-0.079, -0.0243]	1.0000

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.05052	[-0.0796, -0.0215]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.05275	[-0.0791, -0.0264]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.05224	[-0.0789, -0.0256]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.05266	[-0.0815, -0.0238]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.05035	[-0.0759, -0.0248]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.05146	[-0.079, -0.0239]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.1889	[-0.203, -0.175]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.193	[-0.202, -0.184]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.1887	[-0.197, -0.18]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.1925	[-0.201, -0.184]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.1866	[-0.197, -0.176]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.1984	[-0.226, -0.171]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.2096	[-0.245, -0.174]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.2409	[-0.311, -0.17]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	7.129×10^{-4}	$[-4.11 \times 10^{-4}, 0.00184]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	3.055×10^{-4}	$[7.20 \times 10^{-5}, 5.39 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	2.406×10^{-4}	$[-1.33 \times 10^{-4}, 6.14 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	1.358×10^{-4}	$[-3.69 \times 10^{-5}, 3.09 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	1.173×10^{-4}	$[-3.68 \times 10^{-4}, 6.02 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	2.799×10^{-4}	$[-6.96 \times 10^{-5}, 6.29 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	3.193×10^{-4}	$[-3.27 \times 10^{-5}, 6.71 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	1.633×10^{-4}	$[-2.36 \times 10^{-4}, 5.62 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.002075	$[-0.00458, 4.34 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.002434	$[-0.00525, 3.78 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00191	[-0.00581, 0.00199]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.002617	[-0.00626, 0.00103]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.002489	$[-0.00552, 5.44 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.002659	$[-0.00465, -6.66 \times 10^{-4}]$	1.0000

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.001263	[-0.00115, 0.00368]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.003208	[4.90×10^{-4}, 0.00593]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.05848	[0.0553, 0.0616]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.05876	[0.0555, 0.062]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0586	[0.0561, 0.0611]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0584	[0.0552, 0.0616]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.05851	[0.0558, 0.0613]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.05972	[0.0549, 0.0645]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0637	[0.0599, 0.0675]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.07181	[0.0649, 0.0787]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.1907	[0.159, 0.223]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1974	[0.171, 0.224]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.1946	[0.17, 0.22]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1954	[0.17, 0.221]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.1903	[0.162, 0.218]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.2025	[0.163, 0.242]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.2239	[0.174, 0.274]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.2643	[0.182, 0.346]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.05569	[0.0514, 0.06]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.05602	[0.0506, 0.0614]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.05645	[0.0506, 0.0623]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.05565	[0.0497, 0.0615]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0559	[0.05, 0.0618]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.05678	[0.0509, 0.0627]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.06464	[0.0606, 0.0687]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.07486	[0.0684, 0.0813]	1.0000

A.7 id_exact

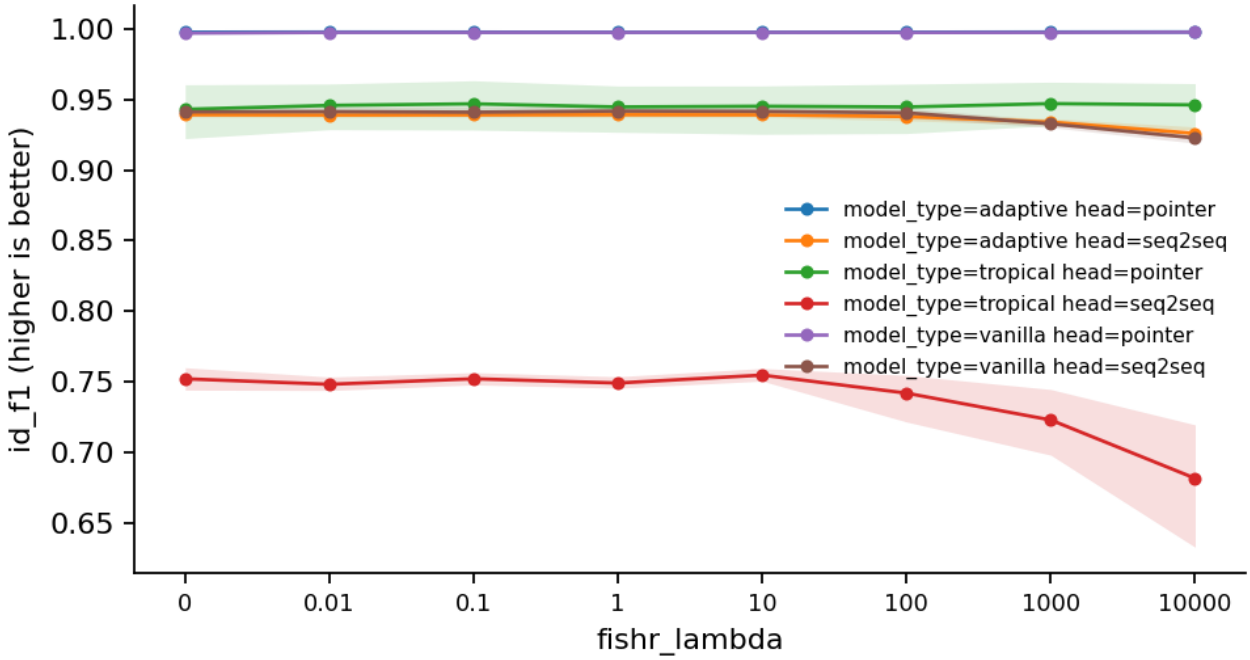


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

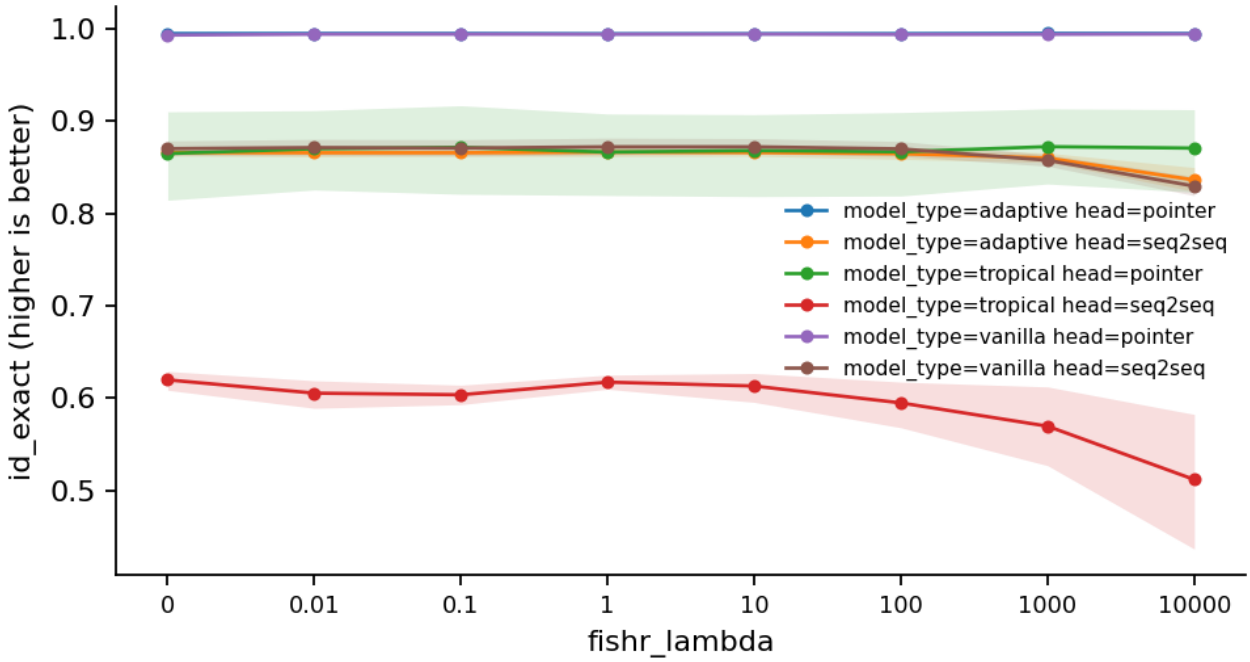


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9949	[0.993, 0.996]	0.00161	5	0
adaptive	pointer	0.01	0.9949	[0.994, 0.995]	6.88×10^{-4}	5	0
adaptive	pointer	0.1	0.9949	[0.994, 0.995]	6.75×10^{-4}	5	0
adaptive	pointer	1	0.9946	[0.994, 0.995]	6.20×10^{-4}	5	0
adaptive	pointer	10	0.9946	[0.994, 0.995]	8.35×10^{-4}	5	0
adaptive	pointer	100	0.9948	[0.994, 0.995]	8.70×10^{-4}	5	0
adaptive	pointer	1000	0.995	[0.994, 0.996]	8.93×10^{-4}	5	0
adaptive	pointer	1e+04	0.9949	[0.995, 0.995]	4.92×10^{-4}	5	0
adaptive	seq2seq	0	0.8658	[0.862, 0.869]	0.00458	5	0
adaptive	seq2seq	0.01	0.8657	[0.861, 0.87]	0.00565	5	0
adaptive	seq2seq	0.1	0.8657	[0.862, 0.869]	0.00492	5	0
adaptive	seq2seq	1	0.8663	[0.862, 0.871]	0.00603	5	0
adaptive	seq2seq	10	0.8659	[0.862, 0.87]	0.00526	5	0
adaptive	seq2seq	100	0.8647	[0.859, 0.871]	0.00797	5	0
adaptive	seq2seq	1000	0.86	[0.855, 0.866]	0.00651	5	0
adaptive	seq2seq	1e+04	0.8362	[0.824, 0.85]	0.016	5	0
tropical	pointer	0	0.8648	[0.814, 0.91]	0.0644	5	0
tropical	pointer	0.01	0.87	[0.826, 0.911]	0.0576	5	0
tropical	pointer	0.1	0.8715	[0.821, 0.917]	0.063	5	0
tropical	pointer	1	0.8664	[0.82, 0.908]	0.0557	5	0
tropical	pointer	10	0.8681	[0.818, 0.907]	0.0559	5	0
tropical	pointer	100	0.8667	[0.819, 0.909]	0.0603	5	0
tropical	pointer	1000	0.8723	[0.832, 0.913]	0.0548	5	0
tropical	pointer	1e+04	0.8707	[0.823, 0.913]	0.0591	5	0
tropical	seq2seq	0	0.6196	[0.609, 0.629]	0.0133	5	0
tropical	seq2seq	0.01	0.6052	[0.589, 0.619]	0.0194	5	0
tropical	seq2seq	0.1	0.6035	[0.593, 0.614]	0.0144	5	0
tropical	seq2seq	1	0.6172	[0.61, 0.625]	0.01	5	0
tropical	seq2seq	10	0.613	[0.595, 0.627]	0.0201	5	0
tropical	seq2seq	100	0.5947	[0.568, 0.617]	0.0309	5	0
tropical	seq2seq	1000	0.5694	[0.527, 0.612]	0.0558	5	0
tropical	seq2seq	1e+04	0.5116	[0.436, 0.582]	0.0964	5	0
vanilla	pointer	0	0.9932	[0.993, 0.994]	5.10×10^{-4}	5	0
vanilla	pointer	0.01	0.9942	[0.994, 0.995]	8.38×10^{-4}	5	0
vanilla	pointer	0.1	0.9942	[0.994, 0.995]	9.07×10^{-4}	5	0
vanilla	pointer	1	0.9941	[0.994, 0.995]	6.76×10^{-4}	5	0
vanilla	pointer	10	0.9942	[0.994, 0.995]	6.27×10^{-4}	5	0
vanilla	pointer	100	0.9939	[0.993, 0.995]	9.35×10^{-4}	5	0
vanilla	pointer	1000	0.994	[0.993, 0.995]	0.00107	5	0
vanilla	pointer	1e+04	0.9943	[0.994, 0.995]	5.70×10^{-4}	5	0
vanilla	seq2seq	0	0.8701	[0.863, 0.879]	0.0101	5	0
vanilla	seq2seq	0.01	0.8713	[0.863, 0.88]	0.0109	5	0
vanilla	seq2seq	0.1	0.8708	[0.862, 0.88]	0.0116	5	0
vanilla	seq2seq	1	0.8721	[0.864, 0.882]	0.0115	5	0
vanilla	seq2seq	10	0.8721	[0.864, 0.881]	0.0108	5	0
vanilla	seq2seq	100	0.8699	[0.862, 0.878]	0.0104	5	0
vanilla	seq2seq	1000	0.8575	[0.851, 0.864]	0.0079	5	0
vanilla	seq2seq	1e+04	0.8296	[0.819, 0.84]	0.0136	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	1.000×10^{-4}	[-0.00168, 0.0017]	1.0000
adaptive	pointer	0.1	-3.000×10^{-4}	[-0.0017, 0.00172]	1.0000
adaptive	pointer	1	-6.000×10^{-4}	[-0.00194, 0.0013]	1.0000
adaptive	pointer	10	-4.000×10^{-4}	[-0.00216, 0.00146]	1.0000
adaptive	pointer	100	-1.000×10^{-4}	[-0.00202, 0.00162]	1.0000
adaptive	pointer	1000	4.000×10^{-4}	[-0.00184, 0.00182]	1.0000
adaptive	pointer	1e+04	1.000×10^{-4}	[-0.00162, 0.0014]	1.0000
adaptive	seq2seq	0.01	-4.000×10^{-4}	[-0.00114, 7.60×10^{-4}]	1.0000
adaptive	seq2seq	0.1	-5.000×10^{-4}	[-0.00104, 8.40×10^{-4}]	1.0000
adaptive	seq2seq	1	6.000×10^{-4}	[-7.20×10^{-4}, 0.00168]	1.0000
adaptive	seq2seq	10	5.000×10^{-4}	[-8.80×10^{-4}, 9.00×10^{-4}]	1.0000
adaptive	seq2seq	100	-3.000×10^{-4}	[-0.00442, 0.0019]	1.0000
adaptive	seq2seq	1000	-0.0066	[-0.00716, -0.00394]	0.4375
adaptive	seq2seq	1e+04	-0.0315	[-0.0385, -0.0193]	0.4375
tropical	pointer	0.01	0.0035	[-0.0019, 0.0125]	1.0000
tropical	pointer	0.1	0.0062	[9.20×10^{-4}, 0.0132]	0.8750
tropical	pointer	1	0.0016	[-0.00504, 0.00872]	1.0000
tropical	pointer	10	0.0051	[-0.00428, 0.00992]	1.0000
tropical	pointer	100	0.0043	[-0.00386, 0.00594]	1.0000
tropical	pointer	1000	0.006	[-0.0013, 0.0176]	1.0000
tropical	pointer	1e+04	0.0097	[-0.00829, 0.017]	1.0000
tropical	seq2seq	0.01	-0.0098	[-0.0203, -0.00878]	0.4375
tropical	seq2seq	0.1	-0.0178	[-0.0224, -0.0113]	0.4375
tropical	seq2seq	1	-0.0044	[-0.0127, 0.0078]	0.8125
tropical	seq2seq	10	-0.0125	[-0.015, 0.00362]	0.6250
tropical	seq2seq	100	-0.0185	[-0.0495, -0.00614]	0.4375
tropical	seq2seq	1000	-0.0452	[-0.0863, -0.0141]	0.4375
tropical	seq2seq	1e+04	-0.0552	[-0.172, -0.0448]	0.4375
vanilla	pointer	0.01	8.000×10^{-4}	[3.00×10^{-4}, 0.00158]	0.6250
vanilla	pointer	0.1	0.0012	[1.40×10^{-4}, 0.0017]	0.6250
vanilla	pointer	1	9.000×10^{-4}	[4.19×10^{-4}, 0.00132]	0.4375
vanilla	pointer	10	0.0012	[4.20×10^{-4}, 0.00158]	0.6250
vanilla	pointer	100	6.000×10^{-4}	[2.00×10^{-5}, 0.00136]	0.6250
vanilla	pointer	1000	6.000×10^{-4}	[6.00×10^{-5}, 0.0016]	0.6250
vanilla	pointer	1e+04	0.0012	[6.00×10^{-4}, 0.00152]	0.4375
vanilla	seq2seq	0.01	0.0019	[-6.00×10^{-5}, 0.00246]	0.9375
vanilla	seq2seq	0.1	0.0018	[-8.80×10^{-4}, 0.00204]	1.0000
vanilla	seq2seq	1	0.0027	[5.00×10^{-4}, 0.00348]	0.7500
vanilla	seq2seq	10	0.0023	[8.60×10^{-4}, 0.00278]	0.6250
vanilla	seq2seq	100	0	[-0.0018, 0.00152]	1.0000
vanilla	seq2seq	1000	-0.0115	[-0.0164, -0.00926]	0.4375
vanilla	seq2seq	1e+04	-0.0435	[-0.0464, -0.0333]	0.4375

Table 31: `id_exact` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.03407	[-0.205, 0.29]	0.8459	0.0324	0.8142
adaptive	seq2seq	-0.5065	[-0.776, -0.364]	0.0019	-0.41	0.0018
tropical	pointer	0.02829	[-0.221, 0.315]	0.8718	0.0286	0.8354
tropical	seq2seq	-0.4088	[-0.652, -0.143]	0.0148	-0.295	0.0314
vanilla	pointer	0	[-0.12, 0.146]	1.0000	-0.00381	0.9834
vanilla	seq2seq	-0.5319	[-0.702, -0.388]	0.0010	-0.421	0.0016

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.1284	[-0.208, -0.0484]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.1242	[-0.196, -0.0521]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.1226	[-0.202, -0.0437]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.1277	[-0.197, -0.0583]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.1262	[-0.196, -0.0563]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.1272	[-0.203, -0.0517]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.1217	[-0.191, -0.0529]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.1236	[-0.197, -0.0497]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.2505	[-0.276, -0.225]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.2661	[-0.3, -0.232]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.2673	[-0.295, -0.24]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.255	[-0.277, -0.232]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.2592	[-0.296, -0.222]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.2752	[-0.315, -0.236]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.2881	[-0.358, -0.218]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.3179	[-0.441, -0.195]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0017	$[-5.49 \times 10^{-4}, 0.00395]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	7.800×10^{-4}	$[1.88 \times 10^{-4}, 0.00137]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	7.400×10^{-4}	$[-2.86 \times 10^{-4}, 0.00177]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	5.200×10^{-4}	$[1.87 \times 10^{-4}, 8.53 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	4.000×10^{-4}	$[-5.78 \times 10^{-4}, 0.00138]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	8.800×10^{-4}	$[-4.25 \times 10^{-5}, 0.0018]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.001	$[2.23 \times 10^{-5}, 0.00198]$	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	5.800×10^{-4}	$[1.38 \times 10^{-4}, 0.00102]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.00426	[-0.0129, 0.00437]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.00562	[-0.0142, 0.00292]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.0051	[-0.0156, 0.00545]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.0058	[-0.0153, 0.00371]	1.0000

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0062	[-0.0145, 0.00212]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00524	[-0.0109, 4.51×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00248	[-0.00279, 0.00775]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00666	[-0.00155, 0.0149]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1291	[0.122, 0.136]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1293	[0.122, 0.137]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1292	[0.123, 0.136]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1283	[0.121, 0.136]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1287	[0.122, 0.135]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1301	[0.12, 0.14]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.135	[0.127, 0.143]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1587	[0.139, 0.178]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.2452	[0.156, 0.334]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.2648	[0.183, 0.346]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.2681	[0.186, 0.35]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.2492	[0.17, 0.328]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.2551	[0.173, 0.337]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.2721	[0.182, 0.362]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.3029	[0.182, 0.423]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.3591	[0.198, 0.52]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1231	[0.111, 0.136]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1229	[0.109, 0.137]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1234	[0.108, 0.138]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.122	[0.108, 0.136]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1221	[0.108, 0.136]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.124	[0.11, 0.138]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1365	[0.126, 0.147]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1647	[0.147, 0.182]	1.0000

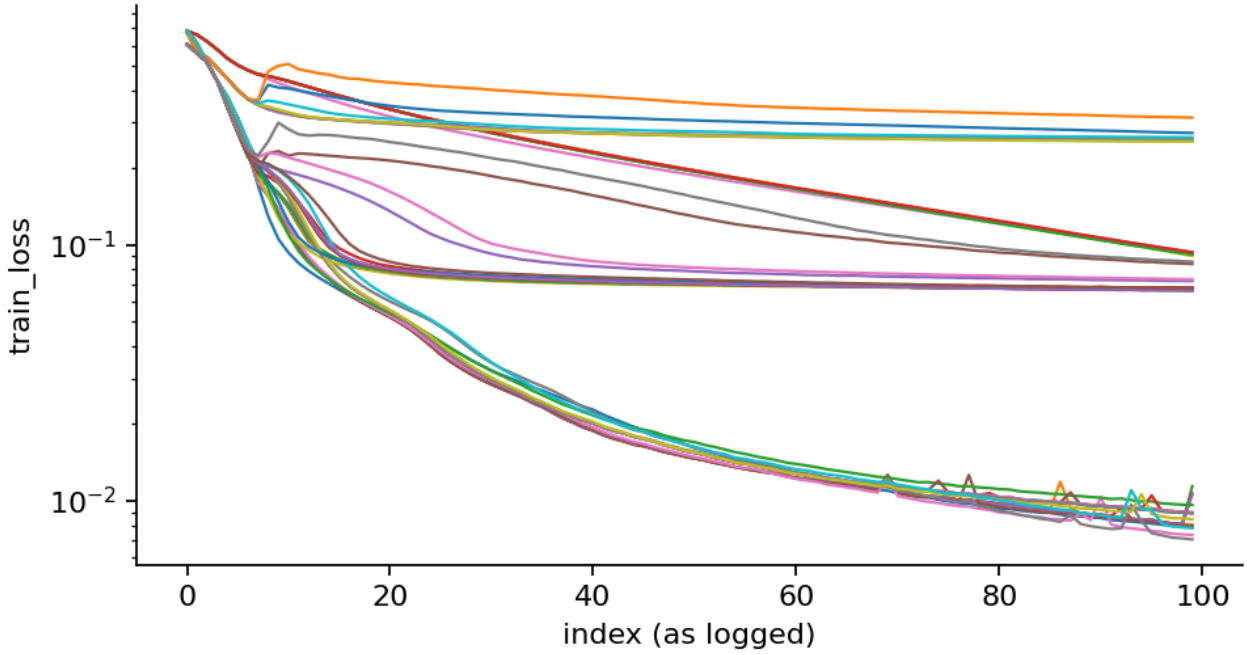


Figure 9: `train_loss`, the mean over seeds per cell.

B Curves

C Configuration

The base configuration every run shares (frozen hash `bd73c304e6b9`); factor levels are in Table 1.

```
task:
  name: QuickselectDataset
  value_range:
  - 1
  - 10
  target_range: null
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
```

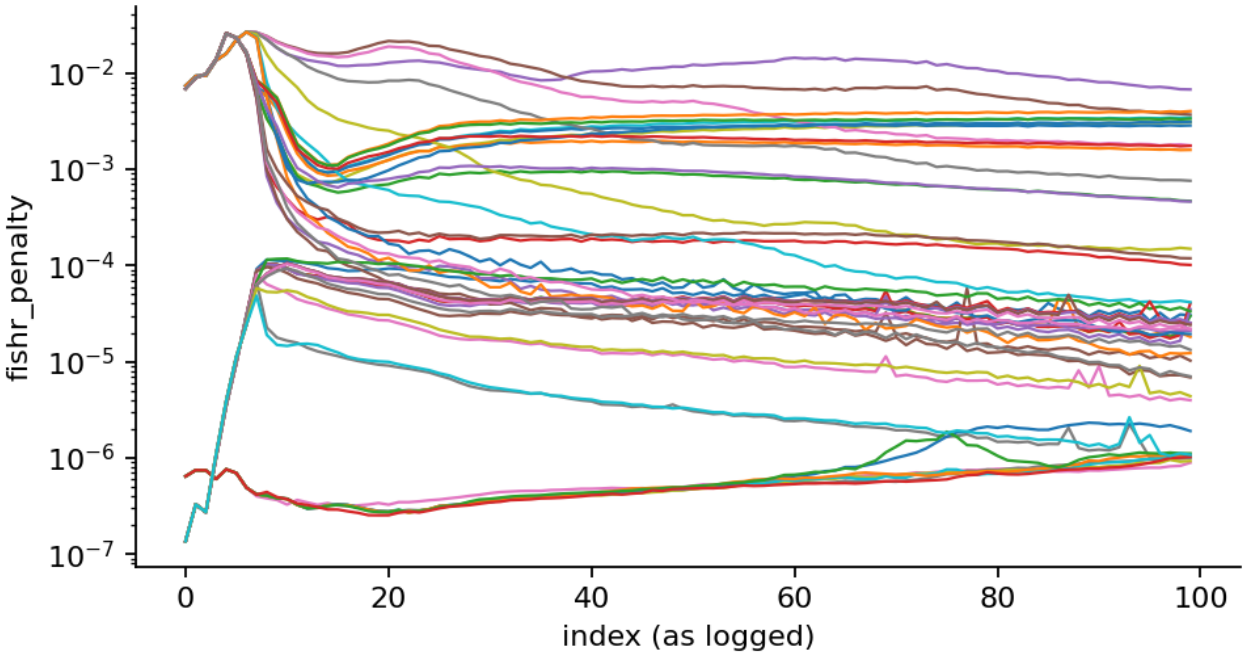


Figure 10: `fishr_penalty`, the mean over seeds per cell.

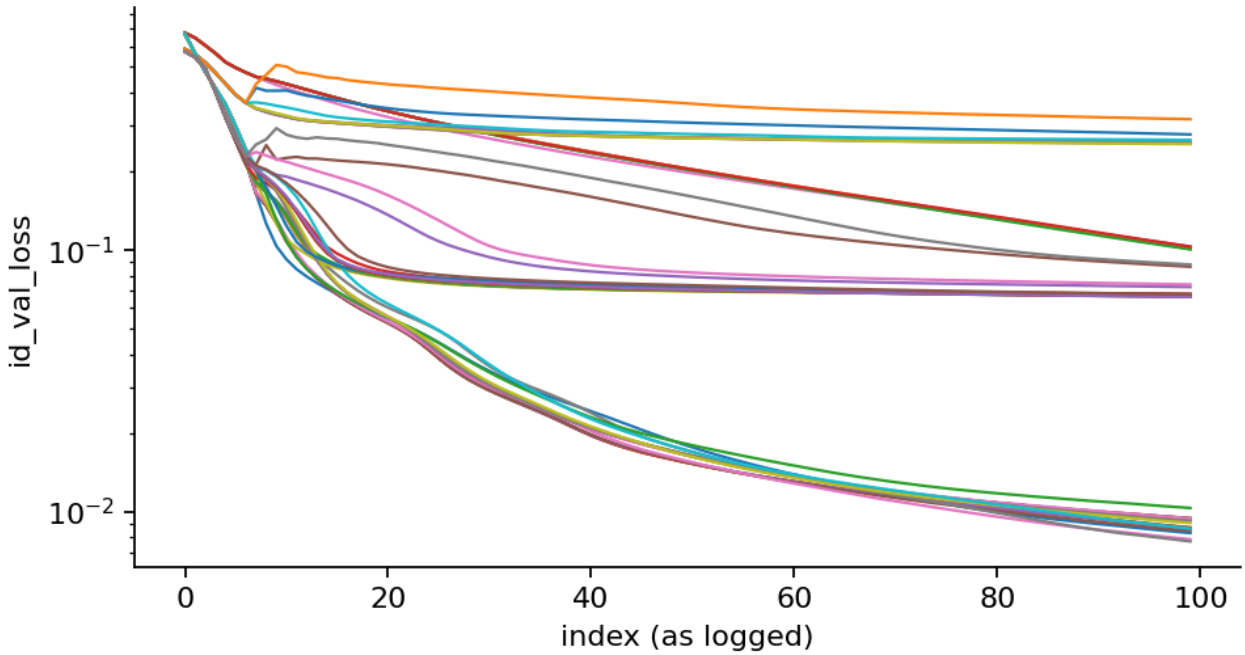


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/Quickselect/results`.